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ZIKI ARC WELDING ELECTRODES

Abbreviations and Symbols

Types of Current Codes

= +	Direct current electrode attached to positive pole
= -	Direct current electrode attached to negative pole
= ±	Direct current electrode attached to either positive or negative pole
~	Alternating current
= + ~	Preferably direct current attached to positive pole, but also alternating current may be used
= - ~	Preferably direct current attached to negative pole, but also alternating current may be used
= ± ~	Preferably direct current attached to either positive or negative pole, but also alternating current may be used

(brackets) - Minimum open circuit voltage when welding using an alternating current

Abbreviations

Rm [MPa]	tensile strength
Rp 0.2 [MPa]	yield strength
A_5 [%]	elongation after rupture (1=5d)
Impact (ISO) [J]	impact value
U	as- welded
S	stress relieved

Notes:

All values listed in this guide are indicative only.
All information contained within this guide is based on careful evaluation.
Regardless, the manufacturer does not assume any liability for their correctness.
Further, the manufacturer reserves the right to change the specifications listed in this guide at any time and without prior notice.

Electrode Brand Name - International Standards

MILD & MEDIUM CARBON STEEL ELECTRODES

ELECTRODE	AWS/ASME SFA	DIN	BS	EN	
Z-610	A5. 1: E 6010	1913: E 43 43 C4	639: E 43 43 C 19	499: E 42 2 C 41	1
Z-611	A5. 1: E 6011	1913: E 43 43 C4	639: E 43 43 C 16	499: E 42 2 C 31	2
Z-6	A5. 1: E 6013	1913: E 43 22 RR6	639: E 43 22 RR 22	499: E 42 0 RR 12	3
Z-9	A5. 1: E 6013	1913: E 42 22 R(C) 3	639: E 43 22 R 12	499: E 42 0 R 12	4
Z-11	A5. 1: E 6013	1913: E 43 22 RR6	639: E 43 22 RR 22	499: E 42 0 RR 12	5
Z-14	A5. 1: E 6013	1913: E 51 22 RR6	639: E 51 22 RR 32	499: E 42 0 RR 12	6
Z-15	A5. 1: E 6013	1913: E 43 22 RR6	639: E 43 22 RR 32	499: E 38 0 RR 12	7
Z-15 (14)	A5. 1: E 6013	1913: E 51 22 RR6	639: E 51 22 RR 32	499: E 42 0 RR 12	8
Z-26	A5. 1: E 6013	1913: E 51 22 R(C)3	639: E 51 22 R 12	499: E 42 0 RC 12	9
Z-16	A5. 1: E 7014	1913: E 51 22 RR(C) 6	639: E 51 22 RR 12	499: E 46 0 RR 32	10
Z-17	A5. 1: E 7024	1913: E 51 22 RR11 140	639: E 51 22 RR 140 35	499: E 42 0 RR 53	11
Z-18	A5. 1: E 7024	1913: E 51 22 RR11 160	639: E 51 22 RR 160 32	499: E 46 0 RR 73	12
Z-19	A5. 1: E 7024	1913: E 51 22 RR11 180	639: E 51 22 RR 190 35	499: E 46 0 RR 73	13
Z-20	A5. 1: E 7024-1	1913: E 51 32 RR11 160	639: E 51 22 RR 160 32	499: E 46 0 RR 73	14
Z-21	A5. 1: E 7024	1913: E 51 22 RR11 180	639: E 51 22 RR 190 35	499: E 46 0 RR 73	15
Z-910	5.5 E-9010-G			2560-A: E-5000Z1	16

LOW ALLOY STEEL ELECTRODES

ELECTRODE	AWS/ASME SFA	DIN	BS	EN	
Z-7	A5. 1: E 7016	1913: E 51 44 B9 H10	639: E 51 44 B 16 HH	499: E 46 3 B 11 H10	17
DIALOGUE	A5. 1: E 7016	1913: E 51 44 B10 H10	639: E 51 44 B 26 HH	499: E 42 3 B 12 H10	18
Z-2	A5. 1: E 7018	1913: E 51 44 B10 H10	639: E 51 44 B 26 HH	499: E 42 3 B 32 H10	19
Z-4	A5. 1: E 7018	1913: E 51 44 B10 H5	639: E 51 44 B 26 H	499: E 46 3 B 32 H5	20
Z-4 AC	A5. 1: E 7018	1913: E 51 44 B10 H5	639: E 51 44 B 26 H	499: E 42 3 B 32 H15	21
Z-55	A5. 1: E 7018-1	8529: EY 46 66 Mn BH5	639: E 51 55 B 120 16 H	499: E 46 5 B 32 H5	22
Z-42	A5. 1: E 7028	1913: E 51 43 B(R)12 160 H5	639: E 51 43 B 160 36 H	499: E 42 3 RB 53 H5	23
Z-47	A5. 1: E 7048	1913: E 51 44 B(R)9 H10	639: E 51 44 B 110 54 HH	499: E 42 3 B 35 H10	24
Z-710	A5. 5: E 7010-A1	-	2493: E Mo C	499: E 42 0 Mo C 41	25
Z-23	A5. 5: E 7018-1	8575: E Mo B 26	2493: E Mo B	1599: E Mo B 32 mod	26
Z- CELLUPIPE	A5. 5: E 8010-G	-	639: E 51 32 C 10	499: E 46 2 1NiMo C 41	27
Z-31 S	A5. 5: E 8010-B2	8575: E CrMo1 B 26	2493: E 1CrMo B	1599: E CrMo1 B 32mod	28
Z-31	A5. 5: E 8010-B2L	8575: E CrMo1 B 26	2493: E 1CrMo L B	1599: E CrMo1 L B 32 mod	29
Z-32	A5. 5: E 8010-C1	8529: E SY 42 77 2Ni B H5	2493: E 2Ni B H	499: E 46 6 2Ni B 42 H5	30
Z-34	A5. 5: E 8010-C3	8529: E Y 46 76 1Ni B H5	2493: E 1Ni B H	-	31
Z-3	A5. 5: E 8010-G	1913: E 51 33 B10 H5	639: E 51 33 B 120 26 H	499: E 50 2 B 32 H5	32
Z-3P	A5.5 E-8018-G	8529: E Y 50 76 Mn1Ni B H5	639: E 51 54 B 120 26 H	503Mo: B 42 H5	33
Z-8	A5. 5: E 8018-G	8529: E SY 50 66 Mn B H5	639: E 51 55 B 120 16 H	499: E 50 6 Mn1Ni B 32 H5	34
Z-33	A5. 5: E 8018-G	8529: E Y 50 32 1NiCu B H5	-	499: E 50 5 B 32 H5	35
Z-3 W	A5. 5: E 8018-W	8575: E CrMo2 B 26	2493: E 2CrMo B	499: E 50 2 B 32 H5	36
Z-41 S	A5. 5: E 9018-B3	8575: E CrMo2 B 26	2493: E 2CrMo L B	1599: E CrMo2 B 32mod	37
Z-41	A5. 5: E 9018-B3L	8529: E SY 55 65 MnMo B H5	2493: E MnMo B H	1599: E CrMo2 L B 32mod	38
Z-46	A5. 5: E 9018-D1	8529: E SY 50 0 1NiMo B H5	-	499: E 50 5 MnMo B 32 H5	39
Z-40 S	A5. 5: E 9018-G	8529: E Y 55 87 2NiMo B H5	2493: E MnNi B H	499: E 50 A 32 H5	40
Z-X70	A5.5 E-9018-G	8529: E Y 62 87 Mn2NiMo B H5	2493: E NiMo B H	499: E 50 A 32 H5	41
Z-48	A5. 5: E 9018-M	8529: E Y 69 76 Mn2NiCrMo B H5	2493: E 2NiMo B H	499: E 50 5 Z B 32 H5	42
Z-50	A5. 5: E 10018-M	8529: E Y 79 76 Mn2Ni1.5CrMo B H5	2493: E 2NiCrMo B H	499: E 50 5 Z B 32 H5	43
Z-52	A5. 5: E 11018-M	8575: E CrMo5 B 26	2493: E 5CrMo B	499: E 50 5 Z B 32 H5	44
Z-53 M	A5. 5: E 12018-M	8575: E CrMo9 B 26	2493: E 9CrMo B	499: E 50 5 Z B 32 H5	45
Z-502	A5. 4: E 502-15			1599: E CrMo5 B 32mod	46
Z-505	A5. 4: E 505-15			1599: E CrMo9 B 32mod	47
Z-91	A5.5 E-9015-B9				48

STAINLESS STEEL ELECTRODES

ELECTRODE	AWS/ASME SFA	DIN	BS	EN	Pg.
Z-307 B	A5.4: E 307-15	-	-	-	49
Z-307 R	A5.4: E 307-16	-	-	-	50
Z-307 HR	A5.4: E 307-16	-	-	-	51
Z-308 B	A5.4: E 308L-15	8556: E 19 9 L B 20+	2926: E 19 9 L B	1600: E 19 9 L B 22	52
Z-308 L	A5.4: E 308L-16	8556: E 19 9 L R 23	2926: E 19 9 L R	1600: E 19 9 L R 12	53
Z-308 LVD	A5.4: E 308L-17	8556: E 19 9 L R 13	2926: E 19 9 L R	1600: E 19 9 L R 11	54
Z-308 LHR	A5.4: E 308L-16	8556: E 19 9 L R 36 150	2926: E 19 9 L R	1600: E 19 9 L R 53 mod	55
Z-308 H	A5.4: E 308-16	8556: E 19 9 R 23	2926: E 19 9 R	1600: E 19 9 R 12	56
Z-309 B	A5.4: E 309L-15	8556: E 23 12 L B 20+	2926: E 23 12 L B	1600: E 23 12 L B 42	57
Z-309 S	A5.4: E 309L-16	8556: E 23 12 L R 23	2926: E 23 12 L R	1600: E 23 12 L R 12	58
Z-309 Cb	A5.4: E 309Cb-16	8556: E 23 12 Nb R 23	2926: E 23 12 Nb R	1600: E 23 12 Nb R 12	59
Z-309 Mo	A5.4: E 309MoL-16	8556: E 23 12 3 R 23	2926: E 23 12 2 R	1600: E 23 12 2 L R 12	60
Z-310	A5.4: E 310-16	8556: E 25 20 R 26	2926: E 25 20 R	1600: E 25 20 R 12 mod	61
Z-312	A5.4: E 312-16	8556: E 29 9 R 23	2926: E 29 9 R	1600: E 29 9 R 12	62
Z-312 S	A5.4: E 312-16	8556: E 29 9 R 23	2926: E 29 9 R	1600: E 29 9 R 12	63
Z-316 B	A5.4: E 316L-15	8556: E 19 123 L B 20+	2926: E 19 12 3 L B	1600: E 19 12 3 L B 22	64
Z-316 L	A5.4: E 316L-16	8556: E 19 123 L R 23	2926: E 19 12 3 L R	1600: E 19 12 3 L R 12	65
Z-316 LVD	A5.4: E 316L-17	8556: E 19 12 3 L R 13	2926: E 19 12 3 L R	1600: E 19 12 3 L R 11	66
Z-316 LHR	A5.4: E 316L-16	8556: E 19 12 3 L R 26 150	2926: E 19 12 3 L R	1600: E 19 12 3 L R 53 mod	67
Z-316 HS	A5.4: E 316-16	8556: E 19 12 3 MPR 23 170	2926: E 19 12 3 RMP	1600: E 19 12 2 R 72	68
Z-317	A5.4: E 317L-16	-	2926: E 19 13 4 L R	1600: E 19 13 4 L R 12 mod	69
Z-318 B	A5.4: E 318-15	8556: E 19 12 3 Nb B 20+	2926: E 19 12 3 Nb B	1600: E 19 12 3 Nb B 22	70
Z-318	A5.4: E 318-16	8556: E 19 12 3 Nb R 23	2926: E 19 12 3 Nb R	1600: E 19 12 3 Nb R 12	71
Z-320	A5.4: E 385-16mod	8556: E 20 25 5 L Cu R 3	-	1600: E 20 25 5 Cu L R 12	72
Z-347 B	A5.4: E 347-15	8556: E 19 9 Nb B 20+	2926: E 19 9 Nb B	1600: E 19 9 Nb B 22	73
Z-347	A5.4: E 347-16	8556: E 19 9 Nb R 23	2926: E 19 9 Nb R	1600: E 19 9 Nb R 12	74
Z-ARM BS	A5.4: E 308Mo-15 mod	-	2926: E 19 9 3 B	-	75
Z-Ni Vi	A5.4: E 308Mo-16 mod	-	2926: E 19 9 3 R	-	76
Z-2209 B	A5.4: E 2209-15	8556: E 22 9 3 L B 20+	-	1600: E 22 9 3 L B 22	77
Z-2209 R	A5.4: E 2209-16	8556: E 22 9 3 L R 23	-	1600: E 22 9 3 L R 12	78
Z-2594 R	A5.4: E 2594-16			1600: E 2594 NLR Z3	79

HARD FACING ELECTRODES

ELECTRODE	AWS/ASME SFA	DIN	BS	EN	Pg.
Z-104	-	8555: E 1 UM 300	-	-	80
Z-105	-	8555: E 1 UM 400	-	-	81
Z-110	-	8555: E 6 UM 55 R	-	-	82
Z-111	-	8555: E 6 UM 55	-	-	83
Z-113	-	8555: E 6 UM 60	-	-	84
Z-114	-	8555: E 6 UM 60	-	-	85
Z-120	-	8555: E 10 UM 60 Z	-	-	86
Z-120 B	-	8555: E 10 UM 60 GR	-	-	87
Z-123	-	8555: E 10 UM 65 GR	-	-	88
Z-125	-	8555: E 10 UM 65 GR	-	-	89
Z-130	A5.13: EFe5-B mod	8555: E 4 UM 60 S	-	-	90
Z-151	A5.13: EFeMn-A	8555: E 7 UM 250 KP	-	-	91
Z-151 NC	A5.13: EFeMn-A mod	8555: E 7 UM 250 KP	-	-	92
Z-152	A5.13: ECoCr-A	-	-	-	93
Z-STEL 6	A5.13: ECoCr-B	8555: E 20 UM 45 CTZ	-	-	94
Z-STEL 12	-	8555: E 20 UM 50 CTZ	-	-	95

NON FERROUS ELECTRODES

ELECTRODE	AWS/ASME SFA	DIN	BS	EN	Pg.
Z-Ni 1	A5. 11: ENi-1	1736: EL-NiTi 3	-	-	96
Z-Ni B	A5. 11: ENiMo-1	1736: EL-NiMo 29	-	-	97
Z-Ni A2	A5. 11: ENiCrFe-2	1736: EL-NiCr 15 FeNb	-	-	98
Z-Ni 182	A5. 11: ENiCrFe-3	1736: EL-NiCr 15 FeMn	-	-	99
Z-Ni 182R1	A5. 11: ENiCrFe-3	1736: EL-NiCr 15 FeMn	-	-	100
Z-Ni 112	A5. 11: ENiCrMo-3	1736: EL-NiCr 20 Mo 9 Nb	-	-	101
Z-Ni 112R	A5. 11: ENiCrMo-3	1736: EL-NiCr 20 Mo 9 Nb	-	-	102
Z-NiCroMo3	A5. 11: ENiCrMo-3	1736: EL-NiCr 20 Mo 9 Nb	-	-	103
Z-NiCroMo4	A5. 11: ENiCrMo-4	1736: EL-NiMo15 Cr 15 W	-	-	104
Z-Ni C	A5. 11: ENiCrMo-5	1736: EL-NiMo15 Cr 15 W mod	-	-	105
Z-Ni 190	A5. 11: ENiCu-7	1736: EL-NiCu 30 Mn	-	-	106
Z-Ni 187	A5. 6: ECuNi	1733: EL-CuNi 30	-	-	107
Z-BRONZE A	A5. 6: ECuSn-A	-	-	-	108
Z-BRONZE	A5. 6: ECuSn-C	1733: EL-CuSn 7	-	-	109
Z-BRONZE M	A5. 6: ECuSn-C	1733: EL-CuSn 7	-	-	110

CAST IRON & CUTTING ELECTRODES

ELECTRODE	AWS/ASME SFA	DIN	BS	EN	Pg.
Z-SuperNi	A5. 15: ENi-CI	8573: E Ni-BG-21	-	-	111
Z-Ni 99	A5. 15: ENi-CI	8573: E Ni-BG-23	-	-	112
Z-Ni 99 NC	A5. 15: ENi-CI	8573: E Ni-BG-23	-	-	113
Z-Ni 55	A5. 15: ENiFe-CI	8573: E NiFe-1-BG-23	-	-	114
Z-GSM	-	8573: E FeC-G-BG-26	-	-	115
Z-GM	A5. 15: ENiCu-B	8573: E NiCu-BG-23	-	-	116
Z-71	-	-	-	-	117
Z-72	-	-	-	-	118
Z-OXYCUT	-	-	-	-	119

FLUX CORED WIRE

ELECTRODE	AWS/ASME SFA	DIN	BS	EN	Pg.
Z-71T1	A5. 20: E71T-1	-	-	-	120
Z-71T5	A5. 20: E71T-5	-	-	-	121

SEMI AUTOMATIC SOLD WIRE

ELECTRODE	AWS/ASME SFA	DIN	BS	EN	Pg.
Z-R2	A5. 18: ER-70S-2	-	-	440: G2 Ti	122
Z-R3	A5. 18: ER-70S-3	8559: SG 1	-	-	123
Z-R6	A5. 18: ER-70S-6	8559: SG 2	2901: A 18	440: G3 Si 1 mod	124
Z-RMo	A5. 28: ER-80S-D2	8559: SG Mo	2901: A 30/A 31	440: G4 Mo	125

Mild & Medium Carbon Steel Electrodes

Z-610

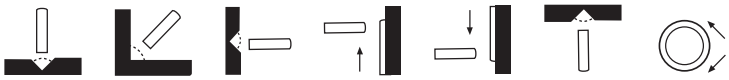
Description and Applications:

- *Cellulose electrode for pipe and general welding*
- *Deep penetration and good X-ray quality*
- *Suitable for welding pipes, pressure vessels, storage tanks, ships*

Specifications:

AWS/ASME SFA A5.1 : E 6010
DIN 1913: E 43 43 C4
BS 639 : E 43 43 C 19
EN 499 : E 42 2 X 41

Welding position



ASME:	1G	2F	2G	3G	3G	4G	5G
ISO 6947	PA	PB	PC	PF	PG	PE	PF/PG

Current

= +

Typical all weld metal composition [wt%]

C	Mn	Si
0.10	0.50	0.20

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A5 [%]	Impact [ISO-V]	Heat Treatment
430	550	25	80 J @-20°-C 45 J @-30°-C	U

Materials to be welded

Unalloyed construction steels	DIN 17 100	St 33 to St 52.3
Boiler and pressure vessel steels	DIN 17 155	HI, HII
Pipe steels	DIN 17 172	StE 210.7 to StE 415.7
	DIN 17 175	St 35.8 to ST 45.8
	DIN 1629	St 35 to St 52.4
	API 5LX	X42 to X63
Fine grained steels	DIN 17 102	StE 255 to StE 355
Shipbuilding steels		A,B,D

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	60-90	1.8	5.0	15.0
1/8	3.25	14	350	80-125	2.9	5.0	15.0
5/32	4.0	14	350	120-180	4.3	5.0	15.0
3/16	5.0	14	350	160-230	6.6	5.0	15.0
1/4	6.0	14	350	220-280	9.5	5.0	15.0

Re-drying not necessary.

Mild & Medium Carbon Steel Electrodes

Z-611

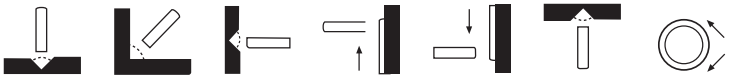
Description and Application:

- *Cellulose electrode for pipe and general welding*
- *Deep penetration and good X-ray quality*
- *Suitable for welding pipes, pressure vessels, storage tanks, ships on both AC and DC*

Specifications:

AWS/ASME SFA A5.1 : E 6011
DIN 1913: E 43 43 C4
BS 639 : E 43 43 C 16
EN 499 : E 42 2 C 31

Welding position



ASME:	1G	2G	2G	3G	3G	4G	5G
ISO 6947	PA	PB	PC	PF	PG	PE	PF/PG

Current

= +	~ (Uo : 70 v)

Typical all weld metal composition [wt%]

C	Mn	Si
0.12	0.50	0.20

Typical all weld metal mechanical properties

Rp 0.2[MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
430	550	25	80 J @-20°C 45 J @-30°C	U

Materials to be welded

Unalloyed construction steels	DIN 17 100	St 33 to St 52.3
Boiler and pressure vessel steels	DIN 17 155	HI, HII
Pipe steels	DIN 17 172	StE 210.7 to StE 415.7
	DIN 17 175	St 35.8 to ST 45.8
	DIN 1629	St 35 to St 52.4
	API 5LX	X42 to X63
Fine grained steels	DIN 17 102	StE 255 to StE 355
Shipbuilding steels		A,B,D

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	60-90	1.8	5.0	15.0
1/8	3.25	14	350	80-125	2.9	5.0	15.0
5/32	4.0	14	350	120-180	4.3	5.0	15.0
3/16	5.0	14	350	160-230	6.6	5.0	15.0
1/4	6.0	14	350	220-280	9.5	5.0	15.0

Re-drying not necessary.

Mild & Medium Carbon Steel Electrodes

Z-6

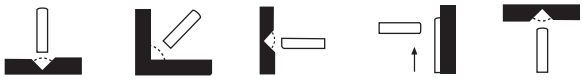
Description and Applications:

- *Rutile coated electrode for welding on uncleaned surfaces like paint, oil or galvanized steel*
- *Usable for pipes, storage tanks and machine parts*

Specifications:

AWS/ASME SFA A5.1 : E 6013
DIN 1913: E 43 22 RR6
BS 639 : E 43 22 RR 22
EN 499 : E 42 0 RR 12

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~	(Uo : 50 V)

Typical all weld metal composition [wt%]

C	Mn	Si
0.07	0.50	0.25

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A5 [%]	Impact [ISO-V]	Heat Treatment
450	550	23	70 J @ 0°C	U

Materials to be welded

Unalloyed construction steels	DIN 17 100	St 33 to St 37.2 to St 44.3
Pipe steels	DIN 17 172	StE 210.7 to StE 360.7
	DIN 1629	St 35 to St 52.4
	API 5LX	X42, X46
Cast steels	DIN 1681	GS 38
Shipbuilding steels		A,B,C,D

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	70-95	1.9	5.5	16.5
1/8	3.25	14	350	100-150	3.0	5.5	16.5
5/32	4.0	14	350	140-200	4.5	5.5	16.5
3/16	5.0	14	350	180-270	7.0	5.5	16.5

Re-drying not necessary.

Mild & Medium Carbon Steel Electrodes

Z-9

Description and Applications:

- *Rutile coated universal electrode for welding general structural steels*
- *Suitable for badly prepared plates of different thicknesses and bad surface conditions*
- *Easy slag release and arc striking*

Specifications:

AWS/ASME SFA A5.1 : E 6013
DIN 1913: E 43 22 R(C)3
BS 639 : E 43 22 R 12
EN 499 : E 42 0 R 12

Welding position

ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~	(Uo : 50 V)

Typical all weld metal composition [wt%]

C	Mn	Si
0.08	0.50	0.30

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
400	520	24	70 J @ 0°C	U

Materials to be welded

Unalloyed construction steels	DIN 17 100	St 33 to St 37.2 to St 44.3
Pipe steels	DIN 17 172	StE 210.7 to StE 360.7
	DIN 1629	St 35 to St 52.4
	API 5LX	X42, X46
Cast steels	DIN 1681	GS 38
Shipbuilding steels		A,B,C,D

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
5/64	2.0	12	300	55-85	1.1	4.0	12.0
3/32	2.5	14	350	70-100	1.9	5.5	16.5
1/8	3.25	14	350	100-150	3.0	5.5	16.5
5/32	4.0	14	350	140-200	4.5	5.5	16.5
3/16	5.0	14/18	350/450	180-230	7.0/9.0	5.5/7.0	16.5/21

Re-drying not necessary.

Mild & Medium Carbon Steel Electrodes

Z-11

Description and Applications:

- *Thickly coated all purpose rutile electrode*
- *Suitable for badly prepared surfaces such as rust and paint and different plate thickness*
- *Very easy slag release and arc striking*

Specifications:

AWS/ASME SFA A5.1 : E 6013
DIN 1913: E 43 22 RR6
BS 639 : E 43 22 RR 22
EN 499 : E 42 0 RR 12

Welding position

ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~	(Uo : 50 V)

Typical all weld metal composition [wt%]

C	Mn	Si
0.08	0.50	0.25

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
440	520	26	70 J @ 0°C	U

Materials to be welded

Unalloyed construction steels	DIN 17 100	St 33, St 37.2 to St 52.3
Pipe steels	DIN 17 172	StE 210.7 to StE 360.7
	DIN 17 175	St 35.8, St 45.8
	DIN 1629	St 35 to St 52.4
Cast steels	DIN 1681	GS 38

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
5/64	2.0	12	300	50-70	1.1	4.0	12.0
3/32	2.5	14	350	60-100	2.0	5.0	15.0
1/8	3.25	14/18	350/450	100-150	3.3/4.2	5.0/6.0	15.0/18.0
5/32	4.0	14/18	350/450	140-200	4.9/6.2	5.0/6.0	15.0/18.0
3/16	5.0	14/18	350/450	180-260	7.7/9.9	5.0/6.0	15.0/18.0

Re-drying not necessary.

Mild & Medium Carbon Steel Electrodes

Z-14

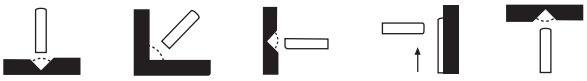
Description and Applications:

- *Thickly coated electrode for general purposes*
- *Exceptionally smooth welding seams, low spatter loss and self detaching slag*

Specifications:

AWS/ASME SFA A5.1 : E 6013
DIN 1913: E 51 22 RR6
BS 639 : E 51 22 RR 32
EN 499 : E 42 0 RR 12

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 50 V)

Typical all weld metal composition [wt%]

C	Mn	Si
0.09	0.55	0.30

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
420	520	24	70 J @ 0°C	U

Materials to be welded

Unalloyed construction steels	DIN 17 100	St 33, St 37.2 to St 52.3
Cast steels	DIN 1681	GS 38, GS 45
Pipe steels	DIN 17 172	StE 210.7 to StE 360.7
	DIN 17 175	St 35.8 to St 45.8
	DIN 1629	St 35 to St 52.4
Shipbuilding steels		A,B,C,D,E

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	70-110	2.1	5.0	15.0
1/8	3.25	18	350/450	100-160	4.8	5.0/6.0	15.0/18.0
5/32	4.0	18	350/450	160-200	7.2	5.0/6.0	15.0/18.0
3/16	5.0	18	450	200-270	10.3	6.0	18.0

Re-drying not necessary.

Mild & Medium Carbon Steel Electrodes

Z-15

Description and Applications:

- *Heavy coated rutile electrode for general purposes*
- *Very even weld in flat and fillet positions*

Specifications:

AWS/ASME SFA A5.1 : E 6013
DIN 1913: E 43 22 RR6
BS 639 : E 43 22 RR 32
EN 499 : E 38 0 RR 12

Welding position

ASME:	1G	2F	2G
ISO 6947	PA	PB	PC

Current

= +	~	(Uo : 50 V)

Typical all weld metal composition [wt%]

C	Mn	Si
0.09	0.60	0.50

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
400	480	24	60 J @ 20°C	U

Materials to be welded

Unalloyed construction steels	DIN 17 100	St 33, St 37.2 to St 52.3
Cast steels	DIN 1681	GS 38, GS 45
Pipe steels	DIN 17 172	StE 210.7 to StE 360.7
	DIN 17 175	St 35.8 to St 45.8
	DIN 1629	St 35 to St 52.4
Shipbuilding steels		A,B,C,D,E

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	70-110	2.1	5.0	15.0
1/8	3.25	18	450	100-160	4.8	6.0	18.0
5/32	4.0	18	450	160-200	7.2	6.0	18.0
3/16	5.0	18	450	200-270	10.3	6.0	18.0

Re-drying not necessary.

Mild & Medium Carbon Steel Electrodes

Z-15 (14)

Description and Applications:

- *Rutile coated electrode for general purposes*
- *Exceptionally smooth welding seams*
- *Suitable for welding thin metal plates*

Specifications:

AWS/ASME SFA A5.1 : E 6013
DIN 1913: E 51 22 RR6
BS 639 : E 51 22 RR 32
EN 499 : E 42 0 RR 12

Welding position

ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 50 V)

Typical all weld metal composition [wt%]

C	Mn	Si
0.07	0.55	0.50

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
420	520	24	50 J @ 0°C	U

Materials to be welded

Unalloyed construction steels	DIN 17 100	St 33, St 37.2 to St 52.3
Cast steels	DIN 1681	GS 38, GS 45
Pipe steels	DIN 17 172	StE 210.7 to StE 360.7
	DIN 17 175	St 35.8 to St 45.8
	DIN 1629	St 35 to St 52.4
Shipbuilding steels		A,B,C,D,E

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	70-110	2.1	5.0	15.0
1/8	3.25	18	350/450	100-160	4.8	5.0/6.0	15.0/18.0
5/32	4.0	18	350/450	160-200	7.2	5.0/6.0	15.0/18.0
3/16	5.0	18	450	200-270	10.30	6.0	18.0

Re-drying not necessary.

Mild & Medium Carbon Steel Electrodes

Z-26

Description and Applications:

- *Rutile Coated all position electrode with a wide range of applications in shipbuilding and structural work*
- *Suitable for badly prepared plates of different thicknesses*

Specifications:

AWS/ASME SFA A5.1 : E 6013
DIN 1913: E 51 22 R(c)3
BS 639 : E 51 22 R 12
EN 499 : E 42 0 RC 12

Welding position

ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 50 V)

Typical all weld metal composition [wt%]

C	Mn	Si
0.07	0.55	0.25

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
470	550	25	60 J @ 0°C	U

Materials to be welded

Unalloyed construction steels	DIN 17 100	St 33, St 37.2 to St 52.3
Boiler and pressure vessel steels	DIN 17 155	HI, HII, 17Mn4, 19Mn5
Pipe steels	DIN 17 172	StE 210.7 to StE 360.7
	DIN 17 175	St 35.8, St 45.8
	DIN 1629	St 35 to St 52.4
Cast steels	DIN 1681	GS 38, GS 45

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
5/64	2.0	12	300	55-70	1.1	4.0/2.0	12.0
3/32	2.5	14	350	70-100	1.9	5.5	16.5
1/8	3.25	14	350	100-150	3.0	5.5	16.5
5/32	4.0	14/18	350/450	140-200	4.5/5.8	5.5/7.0	16.5/21.0
3/16	5.0	14	350/450	160-230	9.0	5.5/7.0	16.5/21.0
1/4	6.0	14	450	240-340	13.0	7.0	21.0

Re-drying not necessary.

Mild & Medium Carbon Steel Electrodes

Z-16

Description and Applications:

- *Thick rutile electrode for fast deposition rate*
- *Focused, stable and penetrating arc*
- *Self detaching slag*
- *Suitable for badly prepared plates of different thicknesses*

Specifications:

AWS/ASME SFA A5.1 : E 7014
DIN 1913: E 51 22 RR(C)6
BS 639 : E 51 22 RR 12
EN 499 : E 46 0 RR 32

Welding position

ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 50 V)

Typical all weld metal composition [wt%]

C	Mn	Si
0.08	0.60	0.25

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
460	540	25	70 J @ 0°C	U

Materials to be welded

Unalloyed construction steels	DIN 17 100	St 33, St 37.2 to St 52.3
Boiler and pressure vessel steels	DIN 17 155	HI, HII, 17Mn4, 19Mn5
Pipe steels	DIN 17 172	StE 210.7 to StE 360.7
	DIN 17 175	St 35.8, St 45.8
	DIN 1629	St 35 to St 52.4
Cast steels	DIN 1681	GS 38, GS 45

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
5/64	2.0	12	300	50-70	1.1	4.0	12.0
3/32	2.5	14	350	60-100	2.0	5.0	15.0
1/8	3.25	14	350	110-150	3.6	5.0	15.0
5/32	4.0	14	350	130-170	4.9	5.0	15.0
3/16	5.0	14	350	190-230	7.5	5.0	15.0

Re-drying not necessary.

Mild & Medium Carbon Steel Electrodes

Z-17

Description and Applications:

- *Rutile coated electrode for welding engineering structures*
- *140% metal recovery*
- *Smooth weld appearance and self detaching slag*

Specifications:

AWS/ASME SFA A5.1 : E 7024
DIN 1913: E 51 22 RR11 140
BS 639 : E 51 22 RR 140 35
EN 499 : E 42 0 RR 53

Welding position

ASME:	1G	2F	2G
ISO 6947	PA	PB	PC

Current

= +	~	(Uo : 50 V)

Typical all weld metal composition [wt%]

C	Mn	Si
0.08	0.50	0.30

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
430	510	25	60 J @ 0°C	U

Materials to be welded

Unalloyed construction steels	DIN 17 100	St 33, St 37.2 to St 52.3
BoilerPlate	DIN 17 155	HI, HII, 17Mn4
Pipe steels	DIN 17 172	StE 210.7 to StE 360.7
	DIN 1629	St 35 to St 52.4
Cast steels	DIN 1681	GS 38
Shipbuilding steels		A,B,C,D

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	60-110	2.4	5.0	15.0
1/8	3.25	18	450	120-150	5.5	6.0	18.0
5/32	4.0	18	450	150-190	7.8	6.0	18.0
3/16	5.0	18	450	200-250	11.5	6.0	18.0

Re-drying: not necessary.

Mild & Medium Carbon Steel Electrodes

Z-18

Description and Applications:

- *Rutile electrode for fillet welds and horizontal V-welds*
- *165% metal recovery with high speed welding*
- *Smooth weld appearance and self detaching slag*

Specifications:

AWS/ASME SFA A5.1 : E 7024
DIN 1913: E 51 22 RR11 160
BS 639 : E 51 22 RR 160 32
EN 499 : E 46 0 RR 73

Welding position

ASME:	1G	2F
ISO 6947	PA	PB

Current

= +	~	(Uo : 50 V)

Typical all weld metal composition [wt%]

C	Mn	Si
0.07	0.70	0.25

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
480	550	26	60 J @ 0°C	U

Materials to be welded

Unalloyed construction steels	DIN 17 100	St 33, St 37.2 to St 52.3
Boiler and pressure vessel steels	DIN 17 155	HI, HII, 17Mn4
Cast steel	DIN 1681	GS 38
Shipbuilding steel		A,B,C,D

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	85-125	3.0	4.5	13.5
1/8	3.25	18	450	130-175	6.5	6.0	18.0
5/32	4.0	18	450	150-240	10.0	6.0	18.0
3/16	5.0	18	450	210-290	15.0	6.0	18.0
1/4	6.0	18	450	290-420	19.5	6.0	18.0

Re-drying: not necessary

Mild & Medium Carbon Steel Electrodes

Z-19

Description and Applications:

- *Rutile electrode for fillet welds and horizontal V-welds*
- *185% metal recovery with high speed welding*
- *Smooth weld appearance and self detaching slag*

Specifications:

AWS/ASME SFA A5.1 : E 7024
DIN 1913: E 51 22 RR11 180
BS 639 : E 51 22 RR 190 35
EN 499 : E 46 0 RR 73

Welding position

ASME:	1G	2F
ISO 6947	PA	PB

Current

= +	~	(Uo : 50 V)

Typical all weld metal composition [wt%]

C	Mn	Si
0.07	0.70	0.25

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
460	550	27	60 J @ 0°C	U

Materials to be welded

Unalloyed construction steels	DIN 17 100	St 33, St 37.2 to St 52.3
Boiler and pressure vessel steels	DIN 17 155	HI, HII, 17Mn4, 19Mn5
Cast steels	DIN 1681	GS 38
Shipbuilding steels		A,B,C,D

Cast steels

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
1/8	3.25	18	450	130-170	6.9	6.0	18.0
5/32	4.0	18	450	150-230	11.2	6.0	18.0
3/16	5.0	18	450	200-350	15.7	6.0	18.0
1/4	6.0	18	450	280-480	22.1	6.0	18.0

Re-drying not necessary.

Mild & Medium Carbon Steel Electrodes

Z-20

Description and Applications:

- *High speed, high deposit (160% metal recovery), rutile electrode*
- *Ideal for very fast fillet welding and for filling horizontal grooves*
- *Fully controlled molten pool, spatter free and self detaching slag*
- *Suitable for welding on moderate and thick plates for buildings, bridges, shipbuilding, etc*

Specifications:

AWS/ASME SFA A5.1 : E 7024-1
DIN 1913: E 51 32 RR11 160
BS 639 : E 51 32 RR 160 32
EN 499 : E 46 0 RR 73

Welding position

ASME:	1G	2F
ISO 6947	PA	PB

Current

= +	~	(Uo : 50 V)

Typical all weld metal composition [wt%]

C	Mn	Si
0.09	0.85	0.45

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
460	550	25	65 J @ 0°C 50 J @ -20°C	U

Materials to be welded

Unalloyed construction steels	DIN 17 100	St 33, St 37.2 to St 52.3
Boiler and pressure vessel steels	DIN 17 155	HI, HII, 17Mn4, 19Mn5
Cast steels	DIN 1681	GS 38
Shipbuilding steels		A,B,C,D

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
1/8	3.25	18	450	130-170	7.2	6.0	18.0
5/32	4.0	18	450	150-230	10.6	6.0	18.0
3/16	5.0	18	450	200-350	16.0	6.0	18.0
1/4	6.0	18	450	280-480	23.8	6.0	18.0

Re-drying not necessary.

Mild & Medium Carbon Steel Electrodes

Z-21

Description and Applications:

- *Rutile electrode for fillet and horizontal V-welds*
- *190% metal recovery with high speed welding*
- *Smooth weld appearance and self detaching slag*

Specifications:

AWS/ASME SFA A5.1 : E 7024
DIN 1913: E 51 22 RR11 180
BS 639 : E 51 22 RR 190 35
EN 499 : E 46 0 RR 73

Welding position

ASME:	1G	2F
ISO 6947	PA	PB

Current

= +	~	(Uo : 50 V)

Typical all weld metal composition [wt%]

C	Mn	Si
0.07	0.70	0.25

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
460	550	27	70 J @ 0°C	U

Materials to be welded

Unalloyed construction steels	DIN 17 100	St 33, St 37.2 to St 52.3
Boiler and pressure vessel steels	DIN 17 155	HI, HII, 17Mn4, 19Mn4
Cast steels	DIN 1681	GS 38
Shipbuilding steels		A,B,C,D

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
1/8	3.25	18	450	130-170	7.2	6.0	18.0
5/32	4.0	18	450	150-230	10.6	6.0	18.0
3/16	5.0	18	450	200-350	16.0	6.0	18.0
1/4	6.0	18	450	280-400	24.3	6.0	18.0

Re-drying not necessary.

Mild & Medium Carbon Steel Electrodes

Z-910

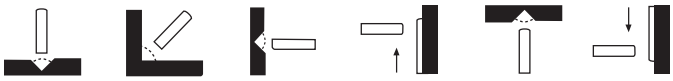
Description and Applications:

- *Cellulose electrode designed for welding in all position including vertical down for high diameter pipelines.*
- *Especially recommended for hot passes, filler and cover layers.*

Specifications:

AWS/ASME SFA 5.5 : E 9010-G
EN ISO 2560-A E50021

Welding position



Typical all weld metal composition [wt%]

C	Mn	Si	Ni	Mo
0.11	1.1	0.55	0.62	0.23

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
690	616	21	79 J @ 0°C

Materials to be welded

API SPEC5L: X65, X70

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	80-125	5	15
1/8	3.25	14	350	110-180	5	15
5/32	4.0	14	350	140-230	5	15

Re-drying not necessary.

Low Alloy Steel Electrodes

Z-7

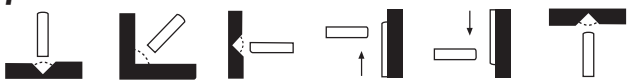
Description and Applications:

- *Basic coated electrode for high quality joining of general construction steels*
- *Very good X-ray properties*
- *Suitable for rool pass welding in narrow joints*
- *Good impact toughness at low temperatures*

Specifications:

AWS/ASME SFA A5.1 : E 7016
DIN 1913: E 51 44 B9 H10
BS 639 : E 51 44 B 16 H
EN 499 : E 46 3 B 11 H10

Welding position



ASME:	1G	2F	2G	3G	3G	4G
ISO 6947	PA	PB	PC	PF	PG	PE

Current

= +	~	(Uo : 70 V)

Typical all weld metal composition [wt%]

C	Mn	Si
0.07	1.00	0.60

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
490	570	26	120 J @ 0°C 100 J @ -20°C 80 J @ -30°C	U

Materials to be welded

General structure steels	DIN 17 100	St 33, St 37.2 to St 52.3
Boiler and pressure vessel steels	DIN 17 155	HI, HII, 17Mn4, 19Mn5
Pipe steels	DIN 17 172	StE 210.7, StE 240.7, StE 290.7, StE 320.7, StE 360.7
	API 5LX	X42, X46, X52, X60
	DIN 1626-1630	St 37.0/4, St 44.0/4, St 52.0/4
Cast steels	DIN 1681	GS 38, GS 45, GS 52
Shipbuilding steels		A,B,C,D,E AH, DH, EH

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	70-80	1.9	5.0	15.0
1/8	3.25	14	350	100-120	3.1	5.0	15.0
5/32	4.0	14	350	140-160	4.7	5.0	15.0
3/16	5.0	18	450	190-210	9.2	6.0	18.0

Re-drying: 2h/300°C

Low Alloy Steel Electrodes

DIALOGUE

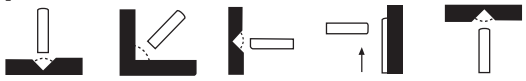
Description and Applications:

- *Double coated basic electrode*
- *Suitable for repair and maintenance work*
- *Excellent for root pass and all positional welding*
- *High impact toughness at low temperatures*

Specifications:

AWS/ASME SFA A5.1 : E 7016
DIN 1913 : E 51 44 B10 H10
BS 639 : E 51 44 B 26 HH
EN 499 : E 42 3 B 12 H10

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 70 V)

Typical all weld metal composition [wt%]

C	Mn	Si
0.06	1.10	0.65

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
420	530	30	110 J @ -20°C 100 J @ -30°C	U

Materials to be welded

General structure steels	DIN 17 100	St 33, St 37.2 to St 52.3
Boiler and pressure vessel steels	DIN 17 155	HI, HII, 17Mn4, 19Mn5
Pipe steels	DIN 17 172	StE 210.7, StE 240.7, StE 290.7, StE 320.7, StE 360.7
	API 5LX	X42, X46, X52, X60
	DIN 1626-1630	St 37.0/4, St 44.0/4, St 52.0/4
Cast steels	DIN 1681	GS 38, GS 45, GS 52
Elevated temperature steels	DIN 17 175	St 35.8, St 45.8
Fine grained steel	DIN 17 102	StE 255 to StE 420
Shipbuilding steels		A,B,C,D,E AH, DH, EH

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	60-90	2.0	4.5	13.5
1/8	3.25	14/18	350/450	95-150	3.4/4.3	4.5	13.5
5/32	4.0	18	350/450	140-190	6.7	5/6	15/18
3/16	5.0	18	450	190-250	10	6	18

Re-drying: 300°C/2h

Low Alloy Steel Electrodes

Z-2

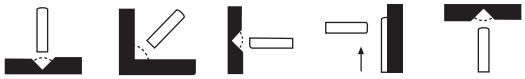
Description and Applications:

- *Basic coated all position welding electrode*
- *Good toughness at low temperatures*
- *Suitable for welding general structural steels*

Specifications:

AWS/ASME SFA A5.1 : E 7018
DIN 1913: E 51 44 B10 H10
BS 639 : E 51 44 B 26 HH
EN 499 : E 42 3 B 32 H10

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~	(Uo : 70 V)

Typical all weld metal composition [wt%]

C	Mn	Si
0.06	1.00	0.35

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
430	510	28	130 J @ -20°C 90 J @ -30°C	U

Materials to be welded

General structure steels	DIN 17 100	St 33, St 37.2 to St 52.3
Boiler and pressure vessel steels	DIN 17 155	HI, HII, 17Mn4, 19Mn5
Pipe steels	DIN 17 172	StE 210.7, StE 240.7, StE 290.7, StE 320.7, StE 360.7
	DIN 1626-1630	St 37.0/4, St 44.0/4, St 52.0/4
Cast steels	DIN 1681	GS 38, GS 45, GS 52
Elevated temperature steels	DIN 17 175	St 35.8, St 45.8
Fine grained steel	DIN 17 102	StE 255 to StE 420
Shipbuilding steels		A,B,C,D,E

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	60-90	2.3	4.5	13.5
1/8	3.25	14/18	350/450	90-140	3.6/4.6	4.5/6.0	13.5/18.0
5/32	4.0	14/18	350/450	140-190	5.3/6.8	4.5/6.0	13.5/18.0
3/16	5.0	18	450	190-240	10.10	6.0	18.0
1/4	6.0	18	450	250-320	14.20	6.0	18.0

Re-drying: 300°C/2h

Low Alloy Steel Electrodes

Z-4

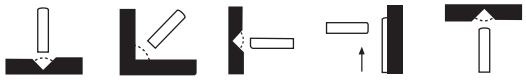
Description and Applications:

- *Basic coated low hydrogen electrode with high ductility*
- *Suitable for welding general construction steels*
- *Very good bead appearrance*
- *Good impact toughness at -30°C*

Specifications:

AWS/ASME SFA A5.1 : E 7018
DIN 1913: E 51 44 B10 H5
BS 639 : E 51 44 B 26 H
EN 499 : E 46 3 B 32 H5

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 70 V)

Typical all weld metal composition [wt%]

C	Mn	Si
0.08	1.00	0.60

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
480	570	26	130 J @ -20°C 100 J @ -30°C	U

Materials to be welded

General structure steels	DIN 17 100	St 33, St 37.2 to St 52.3
Boiler and pressure vessel steels	DIN 17 155	HI, HII, 17Mn4, 19Mn5
Pipe steels	DIN 17 172	StE 210.7, StE 240.7, StE 290.7, StE 320.7, StE 360.7
	DIN 1626-1630	St 37.0/4, St 44.0/4, St 52.0/4
Cast steels	DIN 1681	GS 38, GS 45, GS 52
Elevated temperature steels	DIN 17 175	St 35.8, St 45.8
Fine grained steel	DIN 17 102	StE 255 to StE 420
Shipbuilding steels		A,B,C,D,E

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	75-100	2.6	4.5	13.5
1/8	3.25	14/18	350/450	110-140	3.8/4.9	4.5/6.0	13.5/18.0
5/32	4.0	14/18	350/450	140-190	5.6/7.2	5.0/6.0	15.0/18.0
3/16	5.0	18	450	190-240	10.5	6.0	18.0
1/4	6.0	18	450	230-330	14.3	6.0	18.0

Re-drying: 300°C/2h

Low Alloy Steel Electrodes

Z-4 AC

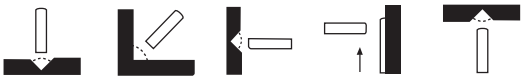
Description and Applications:

- *Basic coated low hydrogen electrode with high ductility*
- *Suitable for welding general construction steels*
- *Very good bead appearrance*
- *Good impact toughness at -30°C*
- *Preferably welded on AC*

Specifications:

AWS/ASME SFA A5.1 : E 7018
DIN 1913: E 51 44 B10 H5
BS 639 : E 51 44 B 26 H
EN 499 : E 42 3 B 32 H5

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 70 V)

Typical all weld metal composition [wt%]

C	Mn	Si
0.09	1.00	0.65

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
450	550	28	130 J @ -20°C 100 J @ -30°C	U

Materials to be welded

General structure steels	DIN 17 100	St 33, St 37.2 to St 52.3
Boiler and pressure vessel steels	DIN 17 155	HI, HII, 17Mn4, 19Mn5
Pipe steels	DIN 17 172	StE 210.7, StE 240.7, StE 290.7, StE 320.7, StE 360.7
	DIN 1626-1630	St 37.0/4, St 44.0/4, St 52.0/4
Cast steels	DIN 1681	GS 38, GS 45, GS 52
Elevated temperature steels	DIN 17 175	St 35.8, St 45.8
Fine grained steel	DIN 17 102	StE 255 to StE 420
Shipbuilding steels		A,B,C,D,E

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	75-100	2.5	4.5	13.5
1/8	3.25	14/18	350/450	110-140	3.7/4.9	4.5/6.0	13.5/18.0
5/32	4.0	14/18	350/450	130-170	5.5/7.1	5.0/6.0	15.0/18.0
3/16	5.0	14/18	350/450	180-240	8.1/10.4	5.0/6.0	15.0/18.0

Re-drying: 300°C/2h

Low Alloy Steel Electrodes

Z-55

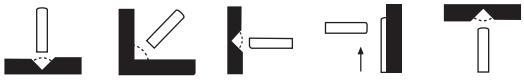
Description and Applications:

- *Low hydrogen basic electrode designed for welding in all positions.*
- *Excellent strength and high toughness down to -46°C. suitable for welding medium and high strength steels.*
- *It can be used in as welded conditions or after heat treatment.*

Specifications:

AWS/ASME SFA A5.1 : E 7018-1
DIN 8529: EY 46 66 MN BH5
EN 499 : E 46 5B 32 H5

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 70 V)

Typical all weld metal composition [wt%]

C	Mn	Si
0.06	1.45	0.45

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
520	600	25	140 J @ -29°C 140 J @ -30°C	
580	500	27	155 J @ -30°C 115J @ -46°C	PWHT 620°C@8hr

Materials to be welded

API SPEC5L: X65, X70, X70

Welding parameters and packaging data

Dia		Length		Current [amps]	Weight packet [kg]	Weight carton [kg]
in	mm	in	mm			
3/32	2.5	14	350	60-90	5	15
1/8	3.25	14	350	110-140	5	15
5/32	4.0	18	450	140-190	6	18
3/16	5.0	17.7	450	190-260	6	18

Low Alloy Steel Electrodes

Z-42

Description and Applications:

- *High efficiency basic electrode (160% metal recovery)*
- *Fast deposition rates and high productivity*
- *Ideal for steel construction, bridges, ship building,*
- *containers and storage tank welding*
- *High porosity resistance*

Specifications:

AWS/ASME SFA A5.1 : E 7028
DIN 1913: E 51 43 B(R)12 160 H5
BS 639 : E 51 43 B 160 36 H
EN 499 : E 42 3 RB 53 H5

Welding position

ASME:	1G	2F
ISO 6947	PA	PB

Current

= +	~ (Uo : 70 V)

Typical all weld metal composition [wt%]

C	Mn	Si
0.05	1.10	0.30

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
540	550	24	120 J @ -20°C 100 J @ -30°C	U

Materials to be welded

Unalloyed construction steels	DIN 17 100	St 33 to St 52.3
Boiler and pressure vessel steels	DIN 17 155	HI, HII, 17Mn4, 19Mn5
Pipe steels	DIN 17 172	StE 210.7, StE 240.7, StE 290.7, StE 320.7, StE 360.7
Cast steels	DIN 1681	GS 38, GS 45, GS 52
Elevated temperature steels	DIN 17 102	StE 255 to StE 420
Fine grained steels	DIN 17 175	St 35.8, St 45.8
Shipbuilding steels		A,B,C,D,E,AH,DH,EH

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
1/8	3.25	18	450	120-190	6.7	5.5	16.5
5/32	4.0	18	450	190-230	10.2	5.5	16.5
3/16	5.0	18	450	230-300	16.0	5.5	16.5
1/4	6.0	18	450	300-350	23.0	5.5	16.5

Re-drying: 300°C/2h

Low Alloy Steel Electrodes

Z-47

Description and Applications:

- *Vertical down welding basic electrode*
- *Recommended for welding in shipbuilding construction*
- *Very little slag and spatter loss*
- *High welding speed*

Specifications:

AWS/ASME SFA A5.1 : E 7048
DIN 1913: E 51 44 B(R)9 H10
BS 639 : E 51 44 B 110 54 HH
EN 499 : E 42 3 B 35 H10

Welding position

ASME:	1G	2F	3G	5G
ISO 6947	PA	PB	PG	PG

Current

= +	~ (Uo : 70 V)

Typical all weld metal composition [wt%]

C	Mn	Si
0.08	1.00	0.60

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
430	530	26	100 J @ -29°C	U

Materials to be welded

General structural steels	DIN 17 100	St 33 to St 52.3
Boiler and pressure vessel steels	DIN 17 155	HI, HII, 17Mn4, 19Mn5
Pipe steels	DIN 17 172	StE 210.7, StE 240.7, StE 290.7, StE 320.7, StE 360.7
Cast steels	DIN 1626-1630	St37.0/4, St44.0/4, St52.0/4
Elevated temperature steels	DIN 1681	GS 38, GS 45, GS 52
Fine grained steels	DIN 17 175	St 35.8, St 45.8
Shipbuilding steels	DIN 17 102	StE 255 to StE 420 A,B,C,D,E,AH,DH,EH

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	70-90	1.7	5.0	15.0
1/8	3.25	14	350/450	90-140	2.9	5.0/6.0	15.0/18.0
5/32	4.0	14	350	135-180	4.4	5.0/6.0	15.0/18.0
3/16	5.0	18	450	175-255	8.9	6.0	18.0

Re-drying: 300°C/2h

Low Alloy Steel Electrodes

Z-710

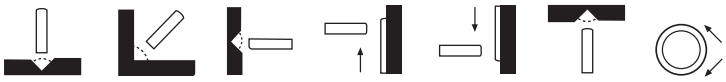
Description and Applications:

- *Cellulose coated electrode*
- *Vertical down welding of high strength pipes*
- *High mechanical properties*

Specifications:

AWS A5.5 : E 7010-A1
BS 2493: E MoC
EN 499 : E 42 0 Mo C 41

Welding position



ASME:	1G	2F	2G	3G	3G	4G	5G
ISO 6947	PA	PB	PC	PF	PG	PE	PF/PG

Current

= +

Typical all weld metal composition [wt%]

C	Mn	Si	Mo
0.10	0.50	0.20	0.50

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Av [ISO-V]	Heat Treatment
500	580	23	70 J @ 0°C	S

Materials to be welded

Unalloyed construction steels	DIN 17 100	St 37 to St 52.3
Boiler and pressure vessel steels	DIN 17 155	HII
Pipe steels	DIN 17 172	StE 210.7 to StE 415.7
	DIN 17 175	St 35.8, St 45.8
	DIN 1629	St 35 to St 52.4
	API 5LX	X42 to X63
Fine grained steels	DIN 17 102	StE 255 to StE 355

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	60-90	1.6	5.0	15.0
1/8	3.25	14	350	90-120	2.8	5.0	15.0
5/32	4.0	14	350	120-160	4.3	5.0	15.0
3/16	5.0	14	350	160-200	6.6	5.0	15.0

Re-drying: not necessary

Low Alloy Steel Electrodes

Z-23

Description and Applications:

- *Basic coated electrode*
- *For welding creep resistant Mo containing low-*
- *alloy steels*
  Withstands service temperatures up to 525°C

Specifications:

AWS/ASME SFA A5.5 : E 7018-A1
DIN 8575: E Mo B 26
BS 2493: E Mo B
EN 1599: E Mo B 32 mod

Welding position

ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~	(Uo : 70 V)

Typical all weld metal composition [wt%]

C	Mn	Si	Mo
0.05	0.70	0.40	0.50

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
410	540	28	140 J @ 20°C	S

Materials to be welded

Boiler and pressure vessel steels	DIN 17 155	14Mn4, 19Mn5, 15Mo3
Pipe steel for elevated temperatures		St 45.8, 15Mo3
Cast steels	DIN 17 245	GS C25, GS 22 Mo4
Elevated temperature steels	DIN 17 175	15Mo3, 16Mo5
Fine grained steels	DIN 17 102	StE 285 to StE 420

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	60-90	2.3	4.5	13.5
1/8	3.25	14	350	90-130	3.5	5.0	15.0
5/32	4.0	14/18	350/450	130-190	5.0/6.0	5.0/6.0	15.0/18.0
3/16	5.0	14	450	180-220	6.0	6.0	18.0

Re-drying: 300°C/2h

Low Alloy Steel Electrodes

Z-CELLUPIPE

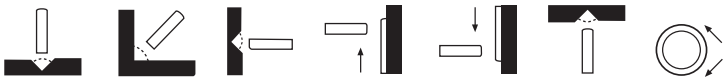
Description and Applications:

- *Cellulosic electrode for vertical down pipe welding*
- *Easy peentration due to forceful arc*
- *Suitable for hot passes in the root, filler and capping passes*
- *Low suscptability to wagon tracks, windows and pinholes*
- *Good X-ray quality*

Specifications:

AWS/ASME SFA A5.5 : E 8010-G
BS 639 : E 51 32 C 10
EN 499 : E 46 2 1NiMo C 41

Welding position



ASME:	1G	2F	2G	3G	3G	4G	5G
ISO 6947	PA	PB	PC	PF	PG	PE	PF/PG

Current

Typical all weld metal composition [wt%]

C	Mn	Si	Ni	Mo
0.10	0.80	0.25	0.45	0.45

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
550	600	23	60 J @ 0°C 50 J @ -20°C	U

Materials to be welded

Pipe steels	DIN 17 172	StE 290.7 to StE 415.7, StE 290.7 TM to StE 445.7 TM
	API 5LX	X42 to X70
Fine grained steels	DIN 17 102	StE 255 to StE 355

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	60-90	1.8	5.0	15.0
1/8	3.25	14	350	80-125	2.9	5.0	15.0
5/32	4.0	18	450	120-180	5.6	5.0	15.0
3/16	5.0	18	450	160-230	8.6	5.0	15.0

Re-drying: not necessary

Low Alloy Steel Electrodes

Z-31 S

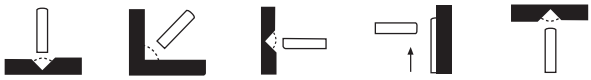
Description and Applications:

- *Basic coated electrode for welding creep resisting steels containing 1% Cr and 0.5% Mo*
- *Service temperatures of maximum 550°C*

Specifications:

AWS/ASME SFA A5.5 : E 8018-B2
DIN 8575 : E CrMo1 B 26
BS 2493 : E 1 CrMo B
EN 1599 : E CrMo 1 B 32 mod

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~	(Uo : 70 V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Mo
0.08	0.80	0.35	1.10	0.50

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Import [ISO-V]	Heat Treatment
540	690	20	120 J @ 20°C	S

Materials to be welded

Elevated temperature steels	DIN 17 155	13CrMo44, 16CrMo44, 15CrMo5, 24CrMo5, 22CrMo44
Quenched and tempered steels	DIN 17 200	25CrMo4
Case hardening steels	DIN 17 210	15Cr3, 16MnCr5, 20MnCr5
Cast steels	DIN 17 245	GS 17CrMo55, GS 22CrMo54, GS 17CrMo55

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	60-100	2.2	4.5	13.5
1/8	3.25	14	350	90-130	3.4	5.0	15.0
5/32	4.0	14/18	350/450	130-180	5.0/6.4	5.0/6.0	15.0/18.0
3/16	5.0	18	450	180-250	10.0	6.0	18.0

Re-drying: 300° C/2h

Low Alloy Steel Electrodes

Z-31

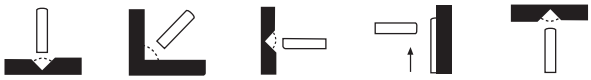
Description and Applications:

- *Basic coated electrode for welding creep resisting steels containing 1% Cr and 0.5% Mo*
- *Service temperatures of maximum 550°C*

Specifications:

AWS/ASME SFA A5.5 : E 8018-B2 L
DIN 8575: E CrMo1 B 26
BS 2493: E 1 CrMo L B
EN 1599 : E CrMo 1 L B 32 mod

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~	(Uo : 70 V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Mo
0.05	0.90	0.50	1.15	0.55

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Import [ISO-V]	Heat Treatment
540	690	20	120 J @ 20°C	S

Materials to be welded

Elevated temperature steels	DIN 17 155	13CrMo44, 16CrMo44, 15CrMo5, 24CrMo5, 22CrMo44
Quenched and tempered steels	DIN 17 200	25CrMo4
Case hardening steels	DIN 17 210	15Cr3, 16MnCr5, 20MnCr5
Cast steels	DIN 17 245	GS 17CrMo55, GS 22CrMo54, GS 17CrMo55

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	60-100	2.2	4.5	13.5
1/8	3.25	14	350	90-130	3.4	5.0	15.0
5/32	4.0	14/18	350/450	130-180	5.0/6.4	5.0/6.0	15.0/18.0
3/16	5.0	18	450	180-250	10.0	6.0	18.0

Re-drying: 300° C/2h

Low Alloy Steel Electrodes

Z-32

Description and Applications:

- *Basic coated electrode for welding low alloy structures subjected to service down to -60ºC*
- *Good corrosion resistance to chloric and sulphuric acid*

Specifications:

AWS/ASME SFA A5.5 : E 8018-C1
DIN 8529: E SY 42 77 2Ni B H5
BS 2493: E 2Ni B H
EN 499 : 46 6 2Ni B 42 H5

Welding position

ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~	(Uo : 70 V)

Typical all weld metal composition [wt%]

C	Mn	Si	Ni
0.05	0.90	0.40	2.15

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Import [ISO-V]	Heat Treatment
520	590	27	110 J @ -59°C	S

Materials to be welded

Fine grained steels	DIN 17 102	StE315 to StE460
Low temperature steels	DIN 17 280	11MnNi53, 13MnNi63, 14NiMn6, 10Ni14

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	80-100	2.2	4.5	13.5
1/8	3.25	14/18	350/450	100-140	3.6/4.	5.0/6.0	15.0/18.0
5/32	4.0	14/18	350/450	130-185	5.4/6.9	5.0/6.0	15.0/18.0
3/16	5.0	18	450	190-250	10.8	6.0	18.0
1/4	6.0	18	450	240-320	15.0	6.0	18.0

Re-drying: 300º C/2h

Low Alloy Steel Electrodes

Z-34

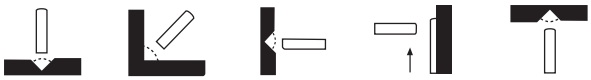
Description and Applications:

- *Low hydrogen basic coated electrode*
- *Suitable for welding of low alloyed, fined grained, high tensile strength steels*

Specifications:

AWS/ASME SFA A5.5 : E 8018-C3
DIN 8529: E Y 46 76 1Ni B H5
BS 2493: E 1Ni B H

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~	(Uo : 70 V)

Typical all weld metal composition [wt%]

C	Mn	Si	Ni	Mo
0.05	0.90	0.35	0.90	0.25

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Import [ISO-V]	Heat Treatment
490	580	28	70 J @ -40°C 40 J @ -60°C	U

Materials to be welded

Fine grained steels	DIN 17 102	StE255 to StE500
Low temperature steels	DIN 17 280	11MnNi53, 13MnNi63, 14NiMn6, 10Ni14

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	75-110	2.3	4.5	13.5
1/8	3.25	14/18	350/450	100-140	3.8/4.9	5.0/6.0	15.0/18.0
5/32	4.0	14/18	350/450	130-200	5.5/7.0	5.0/6.0	15.0/18.0
3/16	5.0	18	450	200-275	11.0	6.0	18.0

Re-drying: 300° C/2h

Low Alloy Steel Electrodes

Z-3

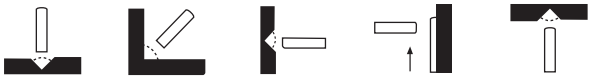
Description and Applications:

- *Basic coated low hydrogen electrode*
- *Suitable for maintenance repairing of agricultural equipment*
- *Electrode produces crack free and tough welded joints*

Specifications:

AWS/ASME SFA A5.5 : E 8018-G
DIN 1913: E 51 33 B10 H5
BS 639 : E 51 33 B 120 26 H
EN 499 : E 50 2 B 32 H5

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~	(Uo : 70 V)

Typical all weld metal composition [wt%]

C	Mn	Si	Mo
0.06	1.30	0.25	0.20

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
500	580	26	150 J @ -20°C	U

Materials to be welded

General structural steels	DIN 17 100	St 52.3
Cast steels	DIN 1681	GS 52
Elevated temperature steels	DIN 17 175	St45.8
Fine grained steels	DIN 17 102	StE 315 to StE 500

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	75-110	2.5	4.5	13.5
1/8	3.25	14/18	350/450	110-140	3.6/4.6	5.0/6.0	15.0/18.0
5/32	4.0	14/18	350/450	140-180	5.1/6.6	5.0/6.0	15.0/18.0
3/16	5.0	14/18	450	190-260	8.0/10.3	6.0	18.0
1/4	6.0	18	450	240-330	15.0	6.0	18.0

Re-drying: 300° C/2h

Low Alloy Steel Electrodes

Z-3P

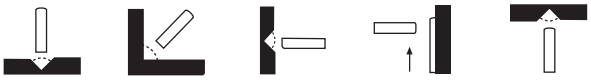
Description and Applications:

- *Low hydrogen welding electrode with basic coating.*
- *Designed for all position welding except vertical down.*
- *Especially recommended for filler and cover layers of pipeline.*

Specifications:

AWS/ASME SFA A5.5 : E 8018-G
EN ISO 2500-A2005 E503MoB42H5

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~	(Uo : 70 V)

Typical all weld metal composition [wt%]

C	Mn	Si	Mo
0.050	1.03	0.38	0.36

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
620	540	26	85 J @ -30°C

Materials to be welded

API SPEC 5L: X46, X52, X56, X60

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	60-90	4.5	13.5
1/8	3.25	14/18	350/450	90-140	5.0/6.0	13.5/18.0
5/32	4.0	18	450	110-190	6.0	18.0

Re-drying: 300° C/2h

Low Alloy Steel Electrodes

Z-8

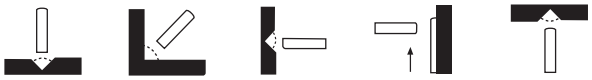
Description and Applications:

- *Basic coated electrode with maximum 1% Ni*
- *Excellent impact toughness at very low temperatures*

Specifications:

AWS/ASME SFA A5.5 : E 8018-G
DIN 8529: E Y 50 76 Mn 1Ni B H5
BS 639 : E 51 54 B 120 26 H
EN 499 : E 50 6 Mn 1Ni B 32 H5

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~	(Uo : 70 V)

Typical all weld metal composition [wt%]

C	Mn	Si	Mo
0.06	1.60	0.30	0.60

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
500	590	30	100 J @ -20°C 90 J @ -40°C 60 J @ -60°C	U

Materials to be welded

General structural steels	DIN 17 100	St 52.3
Cast steels	DIN 1681	GS 52
Elevated temperature steels	DIN 17 175	St45.8
Fine grained steels	DIN 17 102	StE 315 to StE 500
Low temperature steels	DIN 17 280	11MnNi53, 13MnNi63, 14MnNi6

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	60-90	2.1	4.5	13.5
1/8	3.25	14/18	350/450	90-135	3.5/4.5	5.0/6.0	15.0/18.0
5/32	4.0	14/18	350/450	140-180	6.8/8.7	5.0/6.0	15.0/18.0
3/16	5.0	18	450	180-240	10.6	6.0	18.0
1/4	6.0	18	450	260-310	15.1	6.0	18.0

Re-drying: 300° C/2h

Low Alloy Steel Electrodes

Z-33

Description and Applications:

- *Basic coated low hydrogen electrode*
- *Designed for all position welding*
- *Suited for welding high strength fine grained structural steels*
- *High impact toughness at -46°C*

Specifications:

AWS/ASME SFA A5.5 : E 8018-G
DIN 8529: E SY 50 66 Mn B H5
BS 639 : E 51 55 B 120 16 H
EN 499 : E 50 5 B 32 H5

Welding position

ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~	(Uo : 70 V)

Typical all weld metal composition [wt%]

C	Mn	Si
0.08	1.50	0.35

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
530	600	25	135 J @ -29°C 105 J @ -46°C	S

Materials to be welded

General structural steels	DIN 17 100	St 52.3
Cast steels	DIN 1681	GS 52
Elevated temperature steels	DIN 17 175	St45.8
Fine grained steels	DIN 17 102	StE 315 to StE 500
Low temperature steels	DIN 17 280	11MnNi53, 13MnNi63, 14MnNi6

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
5/64	2.0	12	300	50-75	1.2	4.0	12
3/32	2.5	14	350	75-95	2.3	4.5	13.5
1/8	3.25	14/18	350/450	90-140	3.7/4.8	4.5/6.0	13.5/18.0
5/32	4.0	14/18	350/450	135-185	5.8/7.5	5.0/6.0	15.0/18.0
3/16	5.0	18	450	175-255	11.4	6.0	18.0
1/4	6.0	18	450	250-330	16.5	6.0	18.0

Re-drying: 300° C/2h

Low Alloy Steel Electrodes

Z-3 W

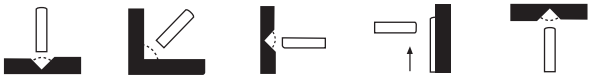
Description and Applications:

- *Basic coated electrode for welding thick structural "weathering" steels of similar composition resistant to atmospheric corrosion*
- *High mechanical properties*

Specifications:

AWS/ASME SFA A5.5 : E 8018-W
DIN 8529: E Y 50 32 1NiCu B H5
EN 499 : E 50 2 B 32 H5

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~	(Uo : 70 V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni	Cu
0.05	1.20	0.35	0.50	0.60	0.50

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
500	600	25	70 J @ -20°C	U

Materials to be welded

Unalloyed construction steels	DIN 17 100	St 37.2 to St 52.3
Boiler and pressure vessel steels	DIN 17 175	HI, HII, 17Mn4, 19Mn5
Pipe steels	API 5LX	X42, X52, X60
Fine grained steels	DIN 17 102	StE 255 to StE 420
Shipbuilding steels		A,B,C,D,E, AH, DH, EH
Weather resistant structural steels		WTSt 37, WTSt 52.3

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	75-110	2.5	4.5	13.5
1/8	3.25	14/18	350/450	110-140	3.7/4.8	4.5/6.0	13.5/18.0
5/32	4.0	14/18	350/450	150-200	5.5/7.1	5.0/6.0	15.0/18.0
3/16	5.0	14/18	450	190-260	8.2/10.5	6.0	18.0

Re-drying: 300° C/2h

Low Alloy Steel Electrodes

Z-41 S

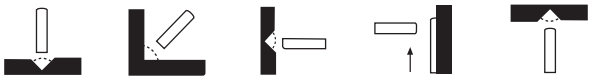
Description and Applications:

- *Basic coated electrode for welding creep resisting steels containing 2% Cr and 1% Mo*
- *Service temperatures of maximum 600°C*

Specifications:

AWS/ASME SFA A5.5 : E 9018-B3
DIN 8575: E CrMo2 B 26
BS 2493: E 2CrMo B
EN 1599: E CrMo2 B 32 mod

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 70 V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Mo
0.09	0.75	0.40	2.25	1.05

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
580	720	18	110 J @ 20°C	S

Materials to be welded

Creep and hydrogen resisting steels	DIN 17 175	10CrMo9 10, 12CrMo9 10, 10CrSiMo V7, 24CrMo10 GS 12CrMo9 10, GS 18CrMo9 10
Cast steels	DIN 17 245	GS 19CrMo9 10

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	70-110	2.2	4.5	13.5
1/8	3.25	14	350	90-130	3.5	5.0	15.0
5/32	4.0	14/18	350/450	130-180	5.1/6.6	5.0/6.0	15.0/18.0
3/16	5.0	18	450	180-250	10.0	6.0	18.0

Re-drying: 300° C/2h

Low Alloy Steel Electrodes

Z-41

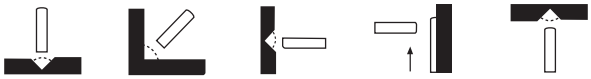
Description and Applications:

- *Basic coated electrode for welding creep resisting steels containing 2% Cr and 1% Mo*
- *Service temperatures of maximum 600°C*

Specifications:

AWS/ASME SFA A5.5 : E 9018-B3 L
DIN 8575: E CrMo2 B 26
BS 2493: E 2CrMo L B
EN 1599: E CrMo2 L B 32 mod

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 70 V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Mo
0.05	0.80	0.40	2.00	1.00

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
580	720	18	110 J @ 20°C	S

Materials to be welded

Creep and hydrogen resisting steels	DIN 17 175	10CrMo9 10, 12CrMo9 10, 10CrSiMo V7, 24CrMo10 GS 12CrMo9 10, GS 18CrMo9 10
Cast steels	DIN 17 245	GS 19CrMo9 10

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	70-110	2.2	4.5	13.5
1/8	3.25	14	350	90-130	3.5	5.0	15.0
5/32	4.0	14/18	350/450	130-180	5.1/6.6	5.0/6.0	15.0/18.0
3/16	5.0	18	450	180-250	10.0	6.0	18.0

Re-drying: 300° C/2h

Low Alloy Steel Electrodes

Z-46

Description and Applications:

- *Basic coated electrode suited for welding high tensile structural steels, working at very low temperatures*
- *Extremely low diffusible hydrogen content*

Specifications:

AWS/ASME SFA A5.5 : E 9018-D1
DIN 8529: E SY 55 65 MnMo B H5
BS 2493: E MnMo B H
EN 499 : E 50 5 MnMo B 32 H5

Welding position

ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 70 V)

Typical all weld metal composition [wt%]

C	Mn	Si	Mo
0.07	1.45	0.40	0.35

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
600	650	23	130 J @ -29°C 80 J @ -51°C	S

Materials to be welded

General structural steels	DIN 17 100	St 33, St 37.2 to St 52.3 DOMEX 400TE, 450, 480
Boiler and pressure vessel steels	DIN 17 155	HI, HII, 17Mn4, 19Mn5
Fine grained steels		USS-T

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	75-100	2.1	5.0	15.0
1/8	3.25	14	350/450	110-140	3.6	5.0/6.0	15.0/18.0
5/32	4.0	14	350/450	150-190	6.0	5.0/6.0	15.0/18.0
3/16	5.0	18	450	190-260	11.2	6.0	18.0

Re-drying: 300° C/2h

Low Alloy Steel Electrodes

Z-40 S

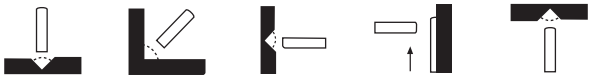
Description and Applications:

- *Basic coated electrode for welding steels such as 4130 and 8630 subjected to heat treating*
- *Welded joints are crack free and tough*

Specifications:

AWS/ASME SFA A5.5 : E 9018-G
DIN 8529: E SY 50 0 1NiMo B H5
EN 499 : E 50 A 32 H5

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Mo	Ni
0.05	1.25	0.40	0.60	0.35	1.20

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
500	600	25	-	S

Materials to be welded

Pipe steels	API 5LX	X70, X 75, X80
Fine grained steels	DIN 17 102	StE420, StE460, StE500
High temperature fine grained steels	DIN 17 102	22NiMoCr37, 17MnMoV64, 15NiCuMoNb5, 20MnMoNi45, 20MnMoNi55

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	75-100	2.1	4.5	13.5
1/8	3.25	14	350	110-140	3.6	5.0	15.0
5/32	4.0	14	350	150-190	5.2	5.0	15.0
3/16	5.0	18	450	190-260	10.3	6.0	18.0

Re-drying: 300º C/2h

Low Alloy Steel Electrodes

Z-X70

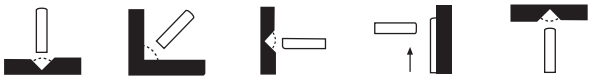
Description and Applications:

- *Low hydrogen basic electrode designed for welding in al positions except vertical down.*
- *Especially recommended for filler and cover layers of high diameter pipelines.*

Specifications:

AWS/ASME SFA A5.5 : E 9018-G
EN 757 E552MnMoB

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 70 V)

Typical all weld metal composition [wt%]

C	Mn	Si	Mo
0.05	1.5	0.32	0.50

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
690	605	25	60 J @ -60°C

Materials to be welded

API SPEC5L: X65, X70,X70

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	60-90	2.25	13.5
1/8	3.25	14	350	90-140	2.5	15
5/32	4.0	14/18	350/450	110-190	2.5	15

Re-drying: 300° C/2h

Low Alloy Steel Electrodes

Z-48

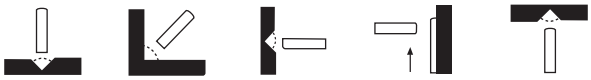
Description and Applications:

- *Basic coated electrode for welding of low temperature, high strength, fine grained structural steels*
- *Good impact toughness at low temperatures*

Specifications:

AWS/ASME SFA A5.5 : E 9018-M
DIN 8529 : E Y 55 87 2NiMo B H5
BS 2493 : E MnNi B H
EN 499 : E 50 5 Z B 32 H5

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~	(Uo : 70 V)

Typical all weld metal composition [wt%]

C	Mn	Si	Ni	Mo
0.05	0.90	0.25	1.75	0.30

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
580	640	24	135 J @ -29°C 105 J @ -51°C 50 J @ -73°C	U

Materials to be welded

General structural steels	DIN 17 100	St 50.2, St 60.2
Boiler and pressure vessel steels	DIN 17 155	15Mo3
Pipe steels	DIN 17 172	StE 210.7, StE 240.7, StE 290.7, StE 320.7, StE 360.7
Cast steels	DIN 1681	GS 60
Elevated temperature steels	DIN 17 175	10Mo3, 16Mo5
Fine grained steels	DIN 17 102	TStE420 to TStE500
High temperature fine grained steels	DIN 17 102	22NiMoCr37, 17MnMoV64, 15 NiCuMoNb5, 20MnMoNi45, 20MnMoNi55

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	70-100	2.1	4.5	13.5
1/8	3.25	14	350	90-140	3.6	5.0	15.0
5/32	4.0	14	350/450	140-190	5.2	5.0/6.0	15.0/18.0
3/16	5.0	18	450	180-250	10.4	6.0	18.0
1/4	6.0	18	450	240-320	14.9	6.0	18.0

Re-drying: 300° C/2h

Low Alloy Steel Electrodes

Z-50

Description and Applications:

- *Basic coated electrode for welding of low temperature, high strength, fine grained structural steels such as navel crafts and submarines.*
- *Particularly low diffusible hydrogen content.*

Specifications:

AWS/ASME SFA A5.5 : E 10018-M
DIN 8529 : E Y 62 87 2Mn2NiMo B H5
BS 2493 : E NiMo B H
EN 499 : E 50 5 Z B 32 H5

Welding position

ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~	(Uo : 70 V)

Typical all weld metal composition [wt%]

C	Mn	Si	Ni	Cr	Mo
0.05	1.65	0.40	2.00	0.20	0.40

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
630	790	22	95 J @ -29°C 65 J @ -51°C 45 J @ -73°C	U

Materials to be welded

General structural steels	DIN 17 100	St 50.2, St 60.2
Boiler and pressure vessel steels	DIN 17 155	15Mo3
Pipe steels	DIN 17 172	StE 210.7, StE 240.7, StE 290.7, StE 320.7, StE 360.7
Cast steels	DIN 1681	GS 60
Elevated temperature steels	DIN 17 175	10Mo3, 16Mo5
Fine grained steels	DIN 17 102	TStE420 to TStE500
High temperature fine grained steels	DIN 17 102	22NiMoCr37, 17MnMoV64, 15 NiCuMoNb5, 20MnMoNi45, 20MnMoNi55

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	70-95	2.3	4.5	13.5
1/8	3.25	14	350/450	90-140	3.7	5.0/6.0	15.0/18.0
5/32	4.0	14	350/450	135-185	5.6	5.0/6.0	15.0/18.0
3/16	5.0	18	450	175-255	11.2	6.0	18.0
1/4	6.0	18	450	250-330	16.2	6.0	18.0

Re-drying: 300° C/2h

Low Alloy Steel Electrodes

Z-52

Description and Applications:

- *Basic coated electrode for welding high strength steels for military applications*
- *Excellent X-ray quality*

Specifications:

AWS/ASME SFA A5.5 : E 11018-M
DIN 8529 : E Y 69 76 Mn2NiCrMo B H5
BS 2493 : E 2NiMo B H
EN 499 : E 50 5 Z B 32 H5

Welding position

ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~	(Uo : 70 V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni	Mo
0.05	1.60	0.30	0.25	2.25	0.40

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
700	780	22	90 J @ -29°C 65 J @ -51°C	U

Materials to be welded

General structural steels	DIN 17 100	St 50.2, St 60.2
Boiler and pressure vessel steels	DIN 17 155	15Mo3
Pipe steels	DIN 17 172	StE 210.7, StE 240.7, StE 290.7, StE 320.7, StE 360.7
Cast steels	DIN 1681	GS 60
Elevated temperature steels	DIN 17 175	10Mo3, 16Mo5
Fine grained steels	DIN 17 102	TStE420 to TStE500
High temperature fine grained steels	DIN 17 102	22NiMoCr37, 17MnMoV64, 15 NiCuMoNb5, 20MnMoNi45, 20MnMoNi55

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	70-95	2.3	4.5	13.5
1/8	3.25	14	350/450	90-140	3.7	5.0/6.0	15.0/18.0
5/32	4.0	14	350/450	135-185	5.6	5.0/6.0	15.0/18.0
3/16	5.0	18	450	175-255	11.2	6.0	18.0
1/4	6.0	18	450	250-330	16.2	6.0	18.0

Re-drying: 300° C/2h

Low Alloy Steel Electrodes

Z-53 M

Description and Applications:

- *Basic coated electrode for welding very high strength steels for military applications*

Specifications:

AWS/ASME SFA A5.5 : E 12018-M
DIN 8529 : E Y 79 76 Mn2Ni1.5CrMo B H5
BS 2493 : E 2NiCrMo B H
EN 499 : E 50 5 Z B 32 H5

Welding position

ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~	(Uo : 70 V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni	Mo
0.06	1.90	0.30	2.20	1.20	0.50

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
800	940	20	60 J @ -29°C 50 J @ -51°C	U

Materials to be welded

General structural steels	DIN 17 100	St 50.2, St 60.2
Boiler and pressure vessel steels	DIN 17 155	15Mo3
Pipe steels	DIN 17 172	StE 210.7, StE 240.7, StE 290.7, StE 320.7, StE 360.7
Cast steels	DIN 1681	GS 60
Elevated temperature steels	DIN 17 175	10Mo3, 16Mo5
Fine grained steels	DIN 17 102	TStE420 to TStE500
High temperature fine grained steels	DIN 17 102	22NiMoCr37, 17MnMoV64, 15 NiCuMoNb5, 20MnMoNi45, 20MnMoNi55

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	65-95	2.1	4.5	13.5
1/8	3.25	14/18	350/450	90-135	3.6	5.0	15.0
5/32	4.0	14/18	350/450	140-185	5.2/6.8	5.0/6.0	15.0/18.0
3/16	5.0	18	450	180-240	10.4	6.0	18.0
1/4	6.0	18	450	240-320	14.9	6.0	18.0

Re-drying: 300° C/2h

Low Alloy Steel Electrodes

Z-502

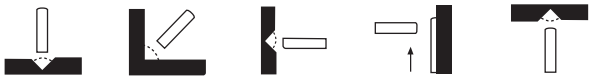
Description and Applications:

- *Basic coated 5% Cr - 0.5% Mo electrode*
- *Service temperature of maximum 600°C*
- *Welding of Cr-Mo alloys of similar composition*
- *Suitable for critical thick joints*

Specifications:

AWS/ASME SFA A5.4 : E 502-15
DIN 8575 : E CrMo5 B 26
BS 2493 : E 5CrMo B
EN 1599 : E CrMo5 B 32 mod

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 70 V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Mo
0.06	0.60	0.35	5.20	0.60

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
490	620	20	80 J @ -20°C	S

Materials to be welded

Boiler and pressure vessel steels	DIN 17 155	12CrMo19 5
Cast steels	DIN 17 245	GS 12CrMo19 5

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	12	300	65-95	2.0	4.0	12.0
1/8	3.25	14	350	90-130	3.6	5.0	15.0
5/32	4.0	14	350	125-180	5.4	5.0	15.0
3/16	5.0	14	350	180-240	8.5	5.0	15.0

Re-drying: 300° C/2h

Low Alloy Steel Electrodes

Z-505

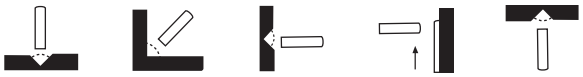
Description and Applications:

- *Basic coated 9% Cr- 1% Mo electrode*
- *Service temperatures of maximum 650^{0}C*

Specifications:

AWS/ASME SFA A5.4 : E 505-15
DIN 8575: E CrMo9 B 26
BS 2493 : E 9CrMO B
EN 1599 : E CrMo9 B 32 mod

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 70V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Mo
0.08	0.75	0.65	9.50	1.00

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
550	700	22	80 J @ 20°C	S

Materials to be welded

Boiler and pressure vessel steels	DIN 17 176	X12 CrMO91
Cast steels	DIN 17 245	GX 12CrMo 10 1

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	12	300	65-105	1.9	4.0	12.0
1/8	3.25	14	350	110-140	3.6	5.0	15.0
5/32	4.0	14	350	150-190	5.4	5.0	15.0
3/16	5.0	14	350	180-230	8.4	5.0	15.0

Re-drying : 300^{0}C/2h

Low Alloy Steel Electrodes

Z-91

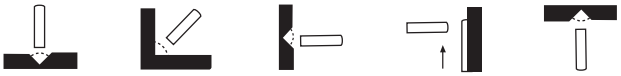
Description and Applications:

Low hydrogen electrode suitable for welding 9%Cr-1%Mo steels, also known as GRADE 91 designed to provide excellent endurance in creep, fatigue, oxidation and corrosion resistance on high temperatures.

Specifications:

AWS/ASME SFA 5.5: E9015-B9

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 70V)

Typical all weld metal composition [wt%]

C	Mn*	Si	Cr	Ni*	Mo	v
0.1	1	0.25	9	0.3	1	0.2

*Mn+Ni<1.5

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm[MPa]	A_5 [%]	Impact [ISO-V]
570	700	18	121J@20°C

Materials to be welded

ASTM
A213 T91 A387 G91 A182/A336 F91
A335 P91 A217 C12A A234 P91
A369 FP91
EN
1.4903 (X10CrMoVNb 91)

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	75-100	4.5	13.5
1/8	3.25	14	350	90-140	6	18
5/32	4.0	14	350	110-190	6	18

Stainless Steel Electrodes

Z-307 B

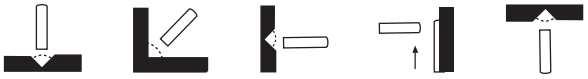
Description and Applications:

- *Basic coated stainless steel electrode*
- *For welding 14% Mn steels, armour plate, steels with unknown composition and hardening steels*
- *Used for buffer layers in hardfacing applications*

Specifications:

AWS/ASME SFA A5.4 : E 307-15

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni	Mo
0.06	4.30	0.40	20.50	9.50	0.80

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
520	620	35	70 J @ 20°C

Materials to be welded

Various steel grades such as:

Non-magnetic austenitic steels
Work hardening austenitic manganese steels
Dissimilar steel grades (CMn-steels to stainless steel)

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	12	300	65-80	1.7	1.5/2.0	9.0/12.0
1/8	3.25	14	300	95-120	3.1	2.0/2.5	12.0/15.0
5/32	4.0	14	300	110-160	4.7	2.0/2.5	12.0/15.0
3/16	5.0	14	300	150-190	6.5	2.0/2.5	12.0/15.0

Re-drying: 300°C2h

Stainless Steel Electrodes

Z-307 R

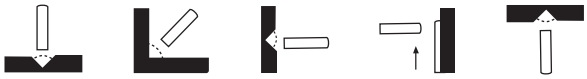
Description and Applications:

- *Rutile coated stainless steel electrode*
- *For welding 14% Mn steels, armour plate, steels with unknown composition and hardening steels*
- *Used for buffer layers in hardfacing applications*

Specifications:

AWS/ASME SFA A5.4 : E 307-16

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~	(Uo : 70V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni	Mo
0.09	4.00	0.80	19.00	10.00	0.90

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
520	620	35	70 J @ 20°C

Materials to be welded

Various steel grades such as:

Non-magnetic austenitic steels
Work hardening austenitic manganese steels
Dissimilar steel grades (CMn-steels to stainless steel)

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	12	300	65-80	1.7	1.5/2.0	9.0/12.0
1/8	3.25	14	350	95-120	3.1	2.0/2.5	12.0/15.0
5/32	4.0	14	350	110-160	4.7	2.0/2.5	12.0/15.0
3/16	5.0	14	350	150-190	6.5	2.0/2.5	12.0/15.0

Re-drying: 300°C2h

Stainless Steel Electrodes

Z-307 HR

Description and Applications:

- *Rutile coated high recovery stainless steel electrode*
- *For welding 14% Mn steels, mixed joints between high alloyed steel, stainless steels and unalloyed base materials*
- *Used for hard to weld steel joints and buffer layers in hardfacing applications*

Specifications:

AWS/ASME SFA A5.4 : E 307-16

Welding position

ASME:	1G	2F	3G
ISO 6947	PA	PB	PF

Current

= +	~ (Uo : 70V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni	Mo
0.08	4.00	0.80	19.50	9.50	1.10

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
-	620	35	100 J @ 20°C

Materials to be welded

Various steel grades such as:

Non-magnetic austenitic steels
Work hardening austenitic manganese steels
Dissimilar steel grades (CMn-steels to stainless steel)

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	65-80	3.2	5.0	5.0
1/8	3.25	18	450	95-120	6.6	6.0	18.0
5/32	4.0	18	450	110-160	10.2	6.0	18.0
3/16	5.0	18	450	150-190	15.0	6.0	18.0

Re-drying : 300°C/2h

Stainless Steel Electrodes

Z-308 B

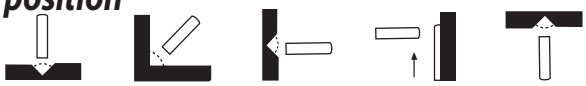
Description and Applications:

- *Basic coated electrode for welding austenitic stainless Cr-Ni steels*
- *Suited for positional welding of thick plates and highly restrained joints*
- *Excellent resistance to cold cracking*
- *Good mechanical properties*

Specifications:

AWS/ASME SFA A5.4 : E 308L-15
DIN 8556: E 19 9 L B 20 +
BS 2926: E 19 9 L B
EN 1600 : 19 9 L B 22

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni
0.03	1.00	0.50	19.00	10.00

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
440	580	40	80 J @ 20°C

Materials to be welded

Extra loiw carbon C<0.035%	DIN 17 440/17 445	X2 CrNi 19 11	1.4306
		X2 CrNiN 18 10	1.4311
Medium carbon C>0.03%	DIN 17 440/17 445	X5 CrNi 18 10	1.4301
		G-X6 CrNi 18 9	1.4308
Ti-, Nb Stabilized	DIN 17 440/17 445	X6 CrNiTi 18 10	1.4541
		X6 CrNiNb 18 10	1.4550
		G-X5 CrNiNb 18 9	1.4552

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
5/64	2.0	12	300	40-55	1.2	1.5/2.0	9.0/12.0
3/32	2.5	12	300	55-80	1.8	1.5/2.0	9.0/12.0
1/8	3.25	14	350	75-110	3.6	2.0/2.5	12.0/15.0
5/32	4.0	14	350	110-150	5.6	2.0/2.5	12.0/15.0
3/16	5.0	18	450	140-180	10.4	6.0	18.0

Stainless Steel Electrodes

Z-308 L

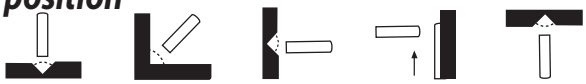
Description and Applications:

- *Rutile-basic coated electrode for welding non stabilized and stabilized corrosion resistant austenitic Cr-Ni steels up to 350^0 C*
- *Smooth metal transfer with both AC and DC*
- *Weld metal beads exhibit fine ripple formation*
- *Focused and stable arc with low spatter loss*
- *Self detachin slag*
- *Moisture resistant formulation producing welds free from porosity*

Specifications:

AWS/ASME SFA A5.4 : E 308L-16
DIN 8556: E 19 9 L R 23
BS 2926: E 19 9 L R
EN 1600 : E 19 9 L R 12

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 50V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni
0.020	0.75	0.80	19.00	10.00

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
450	560	45	70 J @ 20°C

Materials to be welded

Extra loiw carbon C<0.03%	DIN 17 440/17 445	X2 CrNi 19 11	1.4306
		X2 CrNiN 18 10	1.4311
Medium carbon C>0.03%	DIN 17 440/17 445	X5 CrNi 18 10	1.4301
		G-X6 CrNi 18 9	1.4308
Ti-, Nb Stabilized	DIN 17 440/17 445	X6 CrNiTi 18 10	1.4541
		X6 CrNiNb 18 10	1.4550
		G-X5 CrNiNb 18 9	1.4552

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
5/64	2.0	12	300	30-50	1.1	1.5/2.0	9.0/12.0
3/32	2.5	12	300	50-75	1.7	1.5/2.0	9.0/12.0
1/8	3.25	14	350	75-110	3.3	2.0/2.5	12.0/15.0
5/32	4.0	14	350	110-150	4.9	2.0/2.5	12.0/15.0
3/16	5.0	14	450	140-190	7.1	2.0/2.5	12.0/15.0

Re-drying : 300^{0}C/2h

Stainless Steel Electrodes

Z-308 LVD

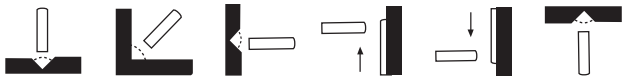
Description and Applications:

- *Ruitle-acid austenitic stainless Cr-Ni steel electrode for welding 304L or equivalent steels in the vertical down position*
- *Designed for welding root passes in grooves with root opening*
- *High corrosion resistance in oxidizing envionments*

Specifications:

AWS/ASME SFA A5.4 : E 308L-17
DIN 8556: E 19 9 L R 13
BS 2926: E 19 9 L R
EN 1600 : E 19 9 L R 11

Welding position



ASME:	1G	2F	2G	3G	3G	4G
ISO 6947	PA	PB	PC	PF	PG	PE

Current

= +	~ (Uo : 50V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni
0.025	0.70	0.70	19.50	10.00

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
440	600	40	70 J @ 20°C

Materials to be welded

Extra loiw carbon C<0.03%	DIN 17 440/17 445	X2 CrNi 19 11	1.4306
		X2 CrNiN 18 10	1.4311
Medium carbon C>0.03%	DIN 17 440/17 445	X5 CrNi 18 10	1.4301
		G-X6 CrNi 18 9	1.4308
Ti-, Nb Stabilized	DIN 17 440/17 445	X6 CrNiTi 18 10	1.4541
		X6 CrNiNb 18 10	1.4550
		G-X5 CrNiNb 18 9	1.4552

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	12	300	60-70	1.4	2.0	12.0
1/8	3.25	12	300	80-100	2.4	2.0	12.0

Re-drying : 300°C/2h

Stainless Steel Electrodes

Z-308 LHR

Description and Applications:

for welding austenitic Cr- Ni steels.

- *Rutile coated high recovery (145%) electrodees.*
- *Used whenever a high deposition rate is required.*
- *Easy slag realease, smooth and finely rippled deposit.*
- *Excellent for fillet and downhand welding.*

Specifications:

AWS/ASME SFA A5.4 : E-308L -16

EN 1600: E 19 9 LR 53 mod

Welding position

ASME:	1G	2F
ISO 6947	PA	PB

Current

= +	~ (Uo : 70V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni
0.025	0.80	0.80	19.50	10.3

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A5 [%]	Impact [ISO-V]
450	580	40	35J@- 105c

Materials to be welded

Extra low carbon C<0.03%	DIN 17 440/17 445	X2 CrNi 19 11	1.4306
Medium carbon C>0.03%	DIN 17 440/17 445	X2 CrNiN 18 10	1.4311
Ti-, Nb Stabilezed	DIN 17 440/ 17 445	X5 CrNi 18 10	1.4301
		G-X6 CrNi 18 9	1.4308
		X6 CrNiTi 18 10	1.4541
		X6 CrNiNb 18 10	1.4550
		G-X5 CrNiNb 18 9	1.4552

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	12	300	75-95	2.5	4.0	12.0
1/8	3.25	14/18	350/450	110-130	5.0/6.40	5.0/6.0	15.0/18.0
5/32	4.0	18	450	135-155	9.3	6.0	18.0
3/16	5.0	18	450	180-210	14.3	6.0	18.0

Re-drying: 300° C/2h

Stainless Steel Electrodes

Z-308 H

Description and Applications:

- *Rutile-basic all position stainless steel electrode*
- *Used for welding low-carbon stainless steels of the type AISI 304 in cases where maximum resistance to intergranular corrosion is required*
- *For welding Titanium and Niobium stabilized stainless steels*
- *Weldable on both AC and DC*

Specifications:

AWS/ASME SFA A5.4 : E 308-16
DIN 8556: E 19 9 R 23
BS 2926 : E 19 9 R
EN 1600: E 19 9 R 12

Welding position

ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 50V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni
0.05	1.00	0.80	20.00	10.00

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
440	580	40	70 J @ 20°C

Materials to be welded

Medium carbon steels	DIN 17 440/17 445	X5 CrNi 18 10	1.4301
		G-X6 CrNi 18 9	1.4308
Ti-, Nb stabilized steels	DIN 17 440/17 445	X6 CrNiTi 18 10	1.4541
		X6 CrNiNb 18 10	1.4550

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
1/16	1.6	12	300	25-40	0.5	1.5/2.0	9.0/12.0
5/64	2.0	12	300	30-50	1.1	1.5/2.0	9.0/12.0
3/32	2.5	12	300	50-75	1.7	1.5/2.0	9.0/12.0
1/8	3.25	14	350	75-110	3.3	2.0/2.5	12.0/15.0
5/32	4.0	14	350	110-150	4.9	2.0/2.5	12.0/15.0
3/16	5.0	14	350	140-190	7.1	2.0/2.5	12.0/15.0

Re-drying : 300°C/2h

Stainless Steel Electrodes

Z-309 B

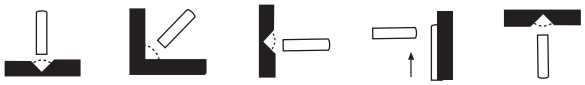
Description and Applications:

- *Basic coated stainless electrode for welding root runs in acid resisting parts of clad steel*
- *Used when overlaying on mild steel or low alloy steels*
- *Joining dissimilar steels (austenitic steels to mild and low alloy steels)*
- *Suitable for welding similar metal joints and austenitic-martensitic steels of the 13% Cr-6% Ni type*
- *Used for buffer layers in hardfacing applicaions*

Specifications:

AWS/ASME SFA A5.4 : E 309L-15
DIN 8556: E 23 12 L B 20+
BS 2926: E 23 12 L B
EN 1600 : E 23 12 L B 42

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni
0.02	0.90	0.40	24.00	12.00

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
470	560	35	70 J @ 20°C

Materials to be welded

Corrosion resisting Clad steel	DIN 17 440/17 465	X2 CrNiN 18 10	1.4311
		X2 CrNi 19 11	1.4306
		X5 CrNi 18 10	1.4301

- Dissimilar metals (mild and low alloyed steel to stainless steel)
- Build-up welding on mild and low alloyed steel

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	12	300	50-80	1.9	1.5/2.0	9.0/12.0
1/8	3.25	14	350	80-110	3.4	2.0/2.5	12.0/15.0
5/32	4.0	14	350	100-150	5.4	2.0/2.5	12.0/15.0
3/16	5.0	14	350	160-220	7.8	2.0/2.5	12.0/15.0

Re-drying : 300°C/2h

Stainless Steel Electrodes

Z-309 S

Description and Applications:

- *Rutile coated low carbon electrode for joining of dissimilar steels (austenitic steels to mild and low alloy steels)*
- *Suitable for welding similar metal joints (23% Cr-12% Ni) and austenitic - martenistic steels of 13% Cr - 6% Ni type*
- *Used for buffer layers in hardfacing applications*
- *Focued and stable arc*
- *Easy slag release*

Specifications:

AWS/ASME SFA A5.4 : E 309L-16
DIN 8556: E 23 12 L R 23
BS 2926 : E 23 12 L R
EN 1600: E 23 12 L R 12

Welding position

ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 50V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni
0.025	0.80	0.80	22.00	12.00

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
420	560	40	60 J @ 20°C

Materials to be welded

Corrosion resisting Clad steel	DIN 17 440/17 465	X2 CrNiN 18 10	1.4311
		X2 CrNi 19 11	1.4306
		X5 CrNi 18 10	1.4301

- Dissimilar metals (mild and low alloyed steel to stainless steel)
- Build-up welding on mild and low alloyed steel

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
5/64	2.0	12	300	30-60	1.2	1.5/2.0	9.0/12.0
3/32	2.5	12	300	40-70	1.8	1.5/2.0	9.0/12.0
1/8	3.25	14	350	70-110	3.4	2.0/2.5	12.0/15.0
5/32	4.0	14	350	110-140	5.1	2.0/2.5	12.0/15.0
3/16	5.0	14	350	140-180	7.3	2.0/2.5	12.0/15.0

Re-drying : 300°C/2h

Stainless Steel Electrodes

Z-309 Cb

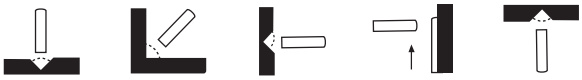
Description and Applications:

- *High CrNiNb alloyed rutile-basic all position buffer electrode*
- *Designed for buffering mild and low alloyed steels for nuclear applications*
- *Used as a buffer electrode in AISI 312 and AISI 347 claddings*
- *Used for the first run when welding 321 or 347 clad steels, prior to completion with 347 type weld metal*
- *Focused and stable arc*
- *Easy slag release*

Specifications:

AWS/ASME SFA A5.4 : E 309Cb-16
DIN 8556: E 23 12 Nb R 23
BS 2926 : E 23 12 Nb R
EN 1600: E 23 12 Nb R 12

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 50V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni	Nb
0.04	0.90	0.80	23.50	13.50	0.80

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
560	680	32	60 J @ 20°C

Materials to be welded

Extra low carbon steels C<0.03%	DIN 17 440	X2 CrNi 19 11	1.4306
		X2 CrNiN 18 10	1.4311
Medium carbon steels C>0.03%	DIN 17 440	X5 CrNi 18 10	1.4301
		G-X6 CrNi 19 9	1.4308
Ti-, Nb stabilized steels	DIN 17 440	X6 CrNiTi 18 10	1.4541
		X6 CrNiNb 18 10	1.4550

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
5/64	2.0	12	300	30-60	1.2	4.0	12.0
3/32	2.5	12	300	40-70	1.8	4.0	12.0
1/8	3.25	14	350	70-110	3.4	5.0	15.0
5/32	4.0	14	350	110-140	5.1	5.0	15.0
3/16	5.0	14	350	140-180	7.3	5.0	15.0

Re-drying: 150°- 200°C/1-2h

Stainless Steel Electrodes

Z-309 Mo

Description and Applications:

- *Rutile-basic coated low carbon all position stainless steel electrode*
- *Suitable for welding high alloyed corrosion resistant Cr-Ni-Mo steels, dissimilar steels (austenitic to ferritic steels)*
- *Depositon of austenitic cladding root runs*
- *Focused and stable arc*
- *Easy slag release*

Specifications:

AWS/ASME SFA A5.4 : E 309MoL-16
DIN 8556: E 23 13 2 R 23
BS 2926: E 23 12 2 R
EN 1600: E 23 12 2 L R 12

Welding position

ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 50V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni	Mo
0.025	0.90	0.80	22.50	12.50	2.80

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
560	550	32	60 J @ 20°C

Materials to be welded

CrNiMo - claddings	DIN 17 440	X2 CrNiMo 17 13 2	1.4404
		X2 CrNiMo 18 14 3	1.4435
		X2 CrNiMoN 17 12 2	1.4406
		X2 CrNiMoN 17 13 3	1.4429
		X5 CrNiMo 17 12 2	1.4401
		X5 CrNiMo 17 13 3	1.4436
		X6 CrNiMoTi17 12 2	1.4571
		X10 CrNiMoTi18 12	1.4573
		X6 CrNiMoNb 17 12 2	1.4580
		X10 CrNiMoNb 18 12	1.4583

- Welsing dissimilar metals: mild steel or low alloyed steel to stainless CrNiMo steel up to max thickness of 12mm
- Build up welding on mild or low alloyed steel

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
5/64	2.0	12	300	30-60	1.2	1.5/2.0	9.0/12.0
3/32	2.5	12	300	40-70	1.8	1.5/2.0	9.0/12.0
1/8	3.25	14	350	70-110	3.5	2.0/2.5	12.0/15.0
5/32	4.0	14	350	110-140	5.2	2.0/2.5	12.0/15.0
3/16	5.0	14	350	140-180	7.5	2.0/2.5	12.0/15.0

Re-drying: 300°C/2h

Stainless Steel Electrodes

Z-310

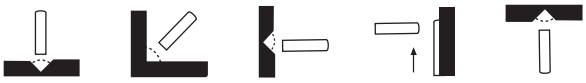
Description and Applications:

- *Basic-rutile coated, fully austenitic all position electrode*
- *Suitable for welding heat resistant Cr and Cr-Ni steels*
- *Oxidation and scalig resistance up to 1200^{0}C*

Specifications:

AWS/ASME SFA A5.4 : E 310-16
DIN 8556 : E 25 20 R 26
BS 2926 : E 25 20 R
EN 1600 : E 25 20 R 12 mod

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 70V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni
0.13	2.10	0.40	26.00	20.00

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
400	580	32	80 J @ 20°C

Materials to be welded

Heat resisting stels — DIN 17 465/43 720

X10 CrAl 24	1.4762
G-X25 CrNiSi 18 9	1.4825
G-X40 CrNiSi 22 9	1.4826
X15 CrNiSi 20 12	1.4828
G-X25 CrNiSi 20 12	1.4832
G-X40 CrNiSi 25 12	1.4837
X15 CrNiSi 25 20	1.4841
G-X15 CrNi 25 20	1.4840
X12 CrNi 25 21	1.4845
G-X40 CrNiSi 25 20	1.4848

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
5/64	2.0	12	300	30-60	1.2	1.5/2.0	9.0/12.0
3/32	2.5	12	300	60-90	1.8	1.5/2.0	9.0/12.0
1/8	3.25	14	350	90-120	3.7	2.0/2.5	12.0/15.0
5/32	4.0	14	350	130-160	5.9	2.0/2.5	12.0/15.0
3/16	5.0	14	350	150-180	8.5	2.0/2.5	12.0/15.0

Re-drying : 300^{0}C/2h

Stainless Steel Electrodes

Z-312

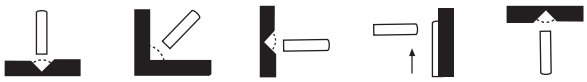
Description and Applications:

- *Rutile-basic coated all position electrode for joining dissimilar steels and surfacing applications*
- *Suitable for welding difficult to weld steels such as: Mn-steel, armour plates*
- *Excellent weldability on both AC and DC*

Specifications:

AWS/ASME SFA A5.4 : E 312-16
DIN 8556: E 29 9 R 23
BS 2926 : E 29 9 R
EN 1600 : E 29 9 R 12

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 50V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni
0.02	0.90	0.80	29.00	9.00

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
500	760	24	-

Materials to be welded

Various steel grades such as:

Hardenable steels including difficult to weld steels
Non-magnetic austenitic steels
Work hardening austenitic manganese steels

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
5/64	2.0	12	300	30-60	1.2	4.0	12.0
3/32	2.5	12	300	60-80	1.9	4.0	12.0
1/8	3.25	14	350	70-110	3.7	5.0	15.0
5/32	4.0	14	350	90-150	5.6	5.0	15.0
3/16	5.0	14	350	150-220	8.3	5.0	15.0

Re-drying :300^{0}C/2h

Stainless Steel Electrodes

Z-312 S

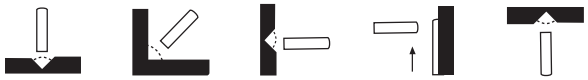
Description and Applications:

- *Rutile-basic coated all position electrode for joining dissimilar steels and surfacing applications*
- *Suitable for welding difficult to weld steels such as: Mn-steel, armour plates*
- *Excellent weldability on both AC and DC*

Specifications:

AWS/ASME SFA A5.4 : E 312-16
DIN 8556: E 29 9 R 23
BS 2926: E 29 9 R
EN 1600 : E 29 9 R 12

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 50V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni
0.11	0.90	0.85	29.00	9.00

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
500	800	24	50 J @ 20°C

Materials to be welded

Various steel grades such as:

Hardenable steels including difficult to weld steels
Non-magnetic austenitic steels
Work hardening austenitic manganese steels

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
5/64	2.0	12	300	30-60	1.1	1.5/2.0	9.0/12.0
3/32	2.5	12	300	60-80	1.7	1.5/2.0	9.0/12.0
1/8	3.25	14	350	70-110	3.3	2.0/2.5	12.0/15.0
5/32	4.0	14	350	90-150	4.9	2.0/2.5	12.0/15.0
3/16	5.0	14	350	150-220	7.0	2.0/2.5	12.0/15.0

Re-drying :300°C/2h

Stainless Steel Electrodes

Z-316 B

Description and Applications:

- *Basic coated low carbon electrode for welding austenitic stainless Cr-Ni-Mo steels*
- *Good mechnical properties*
- *EXcellent position welding properties and penetration*

Specifications:

AWS/ASME SFA A5.4 : E 316L-15
DIN 8556: E 19 12 3 L B 20+
BS 2926: E 19 12 3 L B
EN 1600 : E 19 12 3 L B 22

Welding position

ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni	Mo
0.035	1.00	0.40	18.50	12.00	2.60

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
400	590	35	75 J @ 20°C

Materials to be welded

Extra low carbon C<0.03%	DIN 17 440/17 445	X2 CrNiMo 17 13 2	1.4404
		X2 CrNiMo 18 14 3	1.4435
		X2 CrNiMoN 17 12 2	1.4406
		X2 CrNiMoN 17 13 3	1.4429
Medium carbon C>0.03%	DIN 17 440/17 445	X5 CrNiMo 17 12 2	1.4401
		X5 CrNiMo 17 13 3	1.4436
		G-X6 CrNiMo 18 12	1.4437
		G-X10 CrNiMo 18 9	1.4410
		G-X6 CrNiMo 18 10	1.4408
Ti-, Nb Stabilized	DIN 17 440/17 445	X6 CrNiMoTi 17 12 2	1.4571
		X6 CrNiNb 18 10	1.4550
		G-X5 CrNiNb 18 9	1.4552
		G-X5 CrNiMoNb 18 10	1.4581
		X10 CrNiMoNb 18 12	1.4583

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
5/64	2.0	12	300	40-55	1.2	1.5/2.0	9.0/12.0
3/32	2.5	12	300	55-80	1.8	1.5/2.0	9.0/12.0
1/8	3.25	14	350	75-110	3.6	2.0/2.5	12.0/15.0
5/32	4.0	14	350	110-150	5.6	2.0/2.5	12.0/15.0
3/16	5.0	14	350	140-180	10.4	2.0/2.5	12.0/15.0

Re-drying :300°C/2h

Stainless Steel Electrodes

Z-316 L

Description and Applications:

- *Rutile coated electrode for welding on stabilized corrosion resistant Cr-Ni-Mo steels*
- *Smooth metal transter with both AC and DC*
- *Weld metal beads exhibit fine ripple formatin*
- *Focused and stable arc with low spatter loss*
- *Easy slag release*
- *Moisture resistant formula producing welds free from porosity*

Specifications:

AWS/ASME SFA A5.4 : E 316L-16
DIN 8556: E 19 12 3 L R 23
BS 2926: E 19 12 3 L R
EN 1600 : E 19 12 3 L R 12

Welding position

ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 50V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni	Mo
0.025	0.80	0.80	18.00	11.50	2.80

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
450	580	40	70 J @ 20°C

Materials to be welded

Extra low carbon C<0.03%	DIN 17 440/17 445	X2 CrNiMo 17 13 2	1.4404
		X2 CrNiMo 18 14 3	1.4435
		X2 CrNiMoN 17 12 2	1.4406
		X2 CrNiMoN 17 13 3	1.4429
Medium carbon C>0.03%	DIN 17 440/17 445	X5 CrNiMo 17 12 2	1.4401
		X5 CrNiMo 17 13 3	1.4436
		G-X6 CrNiMo 18 12	1.4437
		G-X10 CrNiMo 18 9	1.4410
		G-X6 CrNiMo 18 10	1.4408
Ti-, Nb Stabilized	DIN 17 440/17 445	X6 CrNiMoTi 17 12 2	1.4571
		X6 CrNiNb 18 10	1.4550
		G-X5 CrNiNb 18 9	1.4552
		G-X5 CrNiMoNb 18 10	1.4581
		X10 CrNiMoNb 18 12	1.4583

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
5/64	2.0	12	300	30-50	1.1	1.5/2.0	9.0/12.0
3/32	2.5	12	300	50-75	1.8	1.5/2.0	9.0/12.0
1/8	3.25	14	350	75-110	3.4	2.0/2.5	12.0/15.0
5/32	4.0	14	350	110-150	5.1	2.0/2.5	12.0/15.0
3/16	5.0	14	350	140-190	7.3	2.0/2.5	12.0/15.0

Re-drying :300°C/2h

Stainless Steel Electrodes

Z-316 LVD

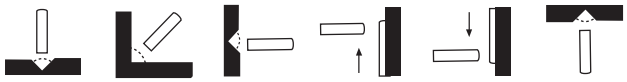
Description and Applications:

- *Rutile-acid austenitic stainless Cr-Ni-Mo steel electrode for welding 316L or equivalent steels in the vertical down position*
- *Designed for welding root passes in grooves with root opening*
- *High corrosion resistance in oxidizing environments*

Specifications:

AWS/ASME SFA A5.4 : E 316L-17
DIN 8556: E 19 12 3 L R 13
BS 2926: E 19 12 3 L R
EN 1600 : E 19 12 3 L R 11

Welding position



ASME:	1G	2F	2G	3G	3G	4G
ISO 6947	PA	PB	PC	PF	PG	PE

Current

= +	~ (Uo : 50V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni	Mo
0.025	0.80	0.70	18.00	11.50	2.70

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
440	600	35	70 J @ 20°C

Materials to be welded

Extra low carbon C<0.03%	DIN 17 440/17 445	X2 CrNiMo 17 13 2	1.4404
		X2 CrNiMo 18 14 3	1.4435
		X2 CrNiMoN 17 12 2	1.4406
		X2 CrNiMoN 17 13 3	1.4429
Medium carbon C>0.03%	DIN 17 440/17 445	X5 CrNiMo 17 12 2	1.4401
		X5 CrNiMo 17 13 3	1.4436
		G-X6 CrNiMo 18 12	1.4437
		G-X10 CrNiMo 18 9	1.4410
		G-X6 CrNiMo 18 10	1.4408
Ti-, Nb Stabilized	DIN 17 440/17 445	X6 CrNiMoTi 17 12 2	1.4571
		X6 CrNiNb 18 10	1.4550
		G-X5 CrNiNb 18 9	1.4552
		G-X5 CrNiMoNb 18 10	1.4581
		X10 CrNiMoNb 18 12	1.4583

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	12	300	60-70	1.4	1.5	9.0
1/8	3.25	12	300	80-100	2.4	1.5	9.0

Re-drying :300°C/2h

Stainless Steel Electrodes

Z-316 LHR

Description and Applications:

- *Rutile coated high recovery (145%) electrode for welding austenitic stainless Cr-Ni-Mo steels*
- *Used whenever a high deposition rate is required*
- *Easy slag rlease, smooth and finely rippled deposit*
- *Excellent for fillet and downhand welding*

Specifications:

AWS/ASME SFA A5.4 : E 316L-16
DIN 8556 : E 19 12 3 L R 26 150
BS 2926 : E 19 12 3 L R
EN 1600 : E 19 12 3 L R 53 mod

Welding position

ASME:	1G	2F
ISO 6947	PA	PB

Current

= +	~ (Uo : 70V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni	Mo
0.025	0.80	0.80	18.00	11.50	2.70

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
470	580	40	70 J @ 20°C

Materials to be welded

Extra low carbon C<0.03%	DIN 17 440/17 445	X2 CrNiMo 17 13 2	1.4404
		X2 CrNiMo 18 14 3	1.4435
		X2 CrNiMoN 17 12 2	1.4406
		X2 CrNiMoN 17 13 3	1.4429
Medium carbon C>0.03%	DIN 17 440/17 445	X5 CrNiMo 17 12 2	1.4401
		X5 CrNiMo 17 13 3	1.4436
		G-X6 CrNiMo 18 12	1.4437
		G-X10 CrNiMo 18 9	1.4410
		G-X6 CrNiMo 18 10	1.4408
Ti-, Nb Stabilized	DIN 17 440/17 445	X6 CrNiMoTi 17 12 2	1.4571
		X6 CrNiNb 18 10	1.4550
		G-X5 CrNiNb 18 9	1.4552
		G-X5 CrNiMoNb 18 10	1.4581
		X10 CrNiMoNb 18 12	1.4583

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	12	300	75-95	2.5	4.0	12.0
1/8	3.25	14/18	350/450	110-130	4.8/6.2	5.0/6.0	15.0/18.0
5/32	4.0	18	450	135-155	9.6	6.0	18.0
3/16	5.0	18	450	180-210	15.1	6.0	18.0

Re-drying :300°C/2h

Stainless Steel Electrodes

Z-316 HS

Description and Applications:

- *High recovery (170%) rutile coated electrode for welding austenitic stainless Cr-Ni-Mo steels*
- *Used whenever easy welding, smooth welding surfaces and a high deposition rate are required*

Specifications:

AWS/ASME SFA A5.4 : E 316-16
DIN 8556: E 19 12 3 MPR 23 170
BS 2926 : E 19 12 3 RMP
EN 1600 : E 19 12 2 R 72

Welding position

ASME:	1G	2F	3G
ISO 6947	PA	PB	PF

Current

= +	~ (Uo : 50V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni	Mo
0.025	0.90	0.70	19.60	11.70	2.80

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
470	580	35	60 J @ 20°C

Materials to be welded

Extra low carbon C<0.03%	DIN 17 440/17 445	X2 CrNiMo 17 13 2	1.4404
		X2 CrNiMo 18 14 3	1.4435
		X2 CrNiMoN 17 12 2	1.4406
		X2 CrNiMoN 17 13 3	1.4429
Medium carbon C>0.03%	DIN 17 440/17 445	X5 CrNiMo 17 12 2	1.4401
		X5 CrNiMo 17 13 3	1.4436
		G-X6 CrNiMo 18 12	1.4437
		G-X10 CrNiMo 18 9	1.4410
		G-X6 CrNiMo 18 10	1.4408
Ti-, Nb Stabilized	DIN 17 440/17 445	X6 CrNiMoTi 17 12 2	1.4571
		X6 CrNiNb 18 10	1.4550
		G-X5 CrNiNb 18 9	1.4552
		G-X5 CrNiMoNb 18 10	1.4581
		X10 CrNiMoNb 18 12	1.4583

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
5/64	2.0	12	300	30-60	1.8	4.0	12.0
3/32	2.5	14	350	65-105	3.3	5.0	15.0
1/8	3.25	14/18	350/450	90-145	5.6/7.2	5.0/6.0	15.0/18.0
5/32	4.0	18	450	145-215	10.9	6.0	18.0
3/16	5.0	18	450	180-270	17.2	6.0	18.0

Re-drying :300°C/2h

Stainless Steel Electrodes

Z-317

Description and Applications:

- *Rutile coated electrode for dissimilar welding*
- *Suitable for difficult to weld steels*
- *Enhanced corrosion resistance applications*

Specifications:

AWS/ASME SFA A5.4 : E 317L-16
BS 2926 : E 19 13 4 L R
EN 1600 : E 19 13 4 L R 12 mod

Welding position

ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 50V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni	Mo
0.025	0.90	0.75	19.00	13.00	3.50

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
500	650	37	100 J @ 20°C

Materials to be welded

Extra low carbon C<0.03%	DIN 17 440/17 445	X2 CrNiMoN 18 18 3	1.4428
		X2 CrNiMoN 17 13 3	1.4429
		X2 CrNiMo 18 14 3	1.4435
		X2 CrNiMo 18 16 4	1.4438

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
5/64	2.0	12	300	35-55	1.2	1.5/2.0	9.0/12.0
3/32	2.5	12	300	40-80	1.8	1.5/2.0	9.0/12.0
1/8	3.25	14	350	80-110	3.4	2.0/2.5	12.0/15.0
5/32	4.0	14	350	110-150	5.2	2.0/2.5	12.0/15.0
3/16	5.0	14	350	150-220	7.4	2.0/2.5	12.0/15.0

Re-drying : 300°C/2h

Stainless Steel Electrodes

Z-318 B

Description and Applications:

- *Basic coated all position electrode for welding Ti, Nb stabilized austenitic stainless Cr-Ni-Mo steels*
- *High reistance to intergranular corrosion up to 400^{0}C*
- *Excellent weldability and easy slag release*

Specifications:

AWS/ASME SFA A5.4 : E 318-15
DIN 8556: E 19 12 3 Nb B 20+
BS 2926: E 19 12 3 Nb B
EN 1600 : E 19 12 3 Nb B 22

Welding position

ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni	Mo	Cb
0.035	1.05	0.35	18.50	11.50	2.75	0.35

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
480	620	35	-

Materials to be welded

Extra low carbon C<0.03%	DIN 17 440/17 445	X2 CrNiMo 17 13 2	1.4404
		X2 CrNiMo 18 14 3	1.4435
		X2 CrNiMoN 17 12 2	1.4406
		X2 CrNiMoN 17 13 3	1.4429
Medium carbon C>0.03%	DIN 17 440/17 445	X5 CrNiMo 17 12 2	1.4401
		X5 CrNiMo 17 13 3	1.4436
		G-X6 CrNiMo 18 12	1.4437
		G-X10 CrNiMo 18 9	1.4410
		G-X6 CrNiMo 18 10	1.4408
Ti-, Nb Stabilized	DIN 17 440/17 445	X6 CrNiMoTi 17 12 2	1.4571
		X6 CrNiNb 18 10	1.4550
		G-X5 CrNiNb 18 9	1.4552
		G-X5 CrNiMoNb 18 10	1.4581
		X10 CrNiMoNb 18 12	1.4583

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
5/64	2.0	12	300	30-50	1.2	1.5/2.0	9.0/12.0
3/32	2.5	12	300	50-70	1.8	1.5/2.0	9.0/12.0
1/8	3.25	14	350	80-100	3.6	2.0/2.5	12.0/15.0
5/32	4.0	14	350	110-140	5.6	2.0/2.5	12.0/15.0
3/16	5.0	14	350	140-180	10.4	2.0/2.5	12.0/15.0

Re-drying :300^{0}C/2h

Stainless Steel Electrodes

Z-318

Description and Applications:

- *Rutile-basic coated all position electrode for welding Ti, Nb stabilized austenitic stainless Cr-Ni-Mo steels*
- *High resistance to intergranular corrosion up to 400^{0}C*
- *Excellent weldability and easy slag release an both AC and DC*

Specifications:

AWS/ASME SFA A5.4 : E 318-16
DIN 8556 : E 19 12 3 Nb R 23
BS 2926 : E 19 12 3 Nb R
EN 1600 : E 19 12 3 Nb R 12

Welding position

ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 50V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni	Mo	Nb
0.025	0.90	0.80	18.00	11.50	1.60	0.40

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
490	600	35	60 J @ 20°C

Materials to be welded

Extra low carbon C<0.03%	DIN 17 440/17 445	X2 CrNiMo 17 13 2	1.4404
		X2 CrNiMo 18 14 3	1.4435
		X2 CrNiMoN 17 12 2	1.4406
		X2 CrNiMoN 17 13 3	1.4429
Medium carbon C>0.03%	DIN 17 440/17 445	X5 CrNiMo 17 12 2	1.4401
		X5 CrNiMo 17 13 3	1.4436
		G-X6 CrNiMo 18 12	1.4437
		G-X10 CrNiMo 18 9	1.4410
		G-X6 CrNiMo 18 10	1.4408
Ti-, Nb Stabilized	DIN 17 440/17 445	X6 CrNiMoTi 17 12 2	1.4571
		X6 CrNiNb 18 10	1.4550
		G-X5 CrNiNb 18 9	1.4552
		G-X5 CrNiMoNb 18 10	1.4581
		X10 CrNiMoNb 18 12	1.4583

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
5/64	2.0	12	300	40-55	1.1	1.5/2.0	9.0/12.0
3/32	2.5	12	300	50-80	1.7	1.5/2.0	9.0/12.0
1/8	3.25	14	350	80-110	3.3	2.0/2.5	12.0/15.0
5/32	4.0	14	350	110-150	4.9	2.0/2.5	12.0/15.0
3/16	5.0	14	350	150-180	7.2	2.0/2.5	12.0/15.0

Re-drying :300^{0}C/2h

Stainless Steel Electrodes

Z-320

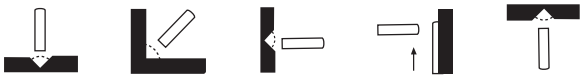
Description and Applications:

- *Rutie-basic fully austenitic stainless steel electrode*
- *High resistance in phosphoric and sulphuric acid applicatons*
- *Smooht metal trasfer on both AC and DC*

Specifications:

AWS/ASME SFA A5.4 : E 385-16 mod
DIN 8556: E 20 25 5 L Cu R 26
EN 1600 : E 20 25 5 Cu L R 12

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 50V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni	Mo	Cu
0.025	1.30	0.85	20.00	25.50	4.70	1.70

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
410	600	40	80 J @ 20°C

Materials to be welded

Fully austenitic NiCrMoCu and CrNiMoCu steel	DIN/SEW 400/410	G-X7 NiCrMoCuNb 25 20	1.4500
		X5 NiCrMoCu Ti 20 18	1.4506
		G-X2 NiCrMoCuN 20 18	1.4531
		G-X2 NiCrMoCuN 25 20	1.4536
		X1 NiCrMoCuN 25 20 5	1.4539
		G-X7 CrNiMoCuNb18 18	1.4585
		X5 NiCrMoCuNb 22 18	1.4586

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	12	300	40-75	1.8	1.5/2.0	9.0/12.0
1/8	3.25	14	350	60-105	3.1	2.0/2.5	12.0/15.0
5/32	4.0	14	350	80-140	4.6	2.0/2.5	12.0/15.0

Re-drying :300°C/2h

Stainless Steel Electrodes

Z-347 B

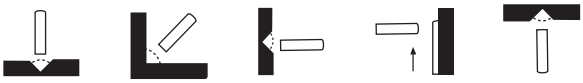
Description and Applications:

- *Basic coated all position electrode for welding Ti, or Nb stabilized austenitic Cr-Ni steels*
- *High reistance to intergranular corrosion up to 400^{0}C*
- *Excellent weldability and easy slag release*

Specifications:

AWS/ASME SFA A5.4 : E 347-15
DIN 8556: E 19 9 Nb B 20+
BS 2926 : E 19 9 Nb B
EN 1600 : E 19 9 Nb B 22

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni	Nb
0.03	1.00	0.50	19.00	10.00	0.35

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
480	620	35	-

Materials to be welded

Ti-, Nb Stabilized	DIN 17 440/17 445	X6 CrNiTi 18 10	1.4541
		X6 CrNiNb 18 10	1.4550
		G-X5 CrNiNb 18 9	1.4552
Non Stabilized	DIN 17 440/17 445	X5 CrNi 18 10	1.4301
		X2 CrNi 19 11	1.4306
		G-X6 CrNi 18 9	1.4308
		G-10 CrNi 18 8	1.4312

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
5/64	2.0	12	300	30-50	1.2	1.5/2.0	9.0/12.0
3/32	2.5	12	300	50-70	1.8	1.5/2.0	9.0/12.0
1/8	3.25	14	350	80-100	3.6	2.0/2.5	12.0/15.0
5/32	4.0	14	350	110-140	5.6	2.0/2.5	12.0/15.0
3/16	5.0	14	350	140-180	10.4	6.0	18.0

Re-drying :300^{0}C/2h

Stainless Steel Electrodes

Z-347

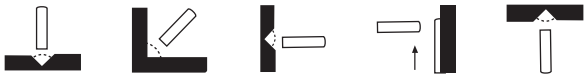
Description and Applications:

- *Rutile-basic coated all position electrode for welding Ti or Nb stabilized austenitic Cr-Ni steels*
- *High resistance to intergranular corrosion up to 400^{0}C*
- *Excellent weldability and slag release an both AC and DC*

Specifications:

AWS/ASME SFA A5.4 : E 347-16
DIN 8556: E 19 9 Nb R 23
BS 2926 : E 19 9 Nb R
EN 1600 : E 19 9 Nb R 12

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 50V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni	No
0.03	0.80	0.80	19.00	10.00	0.40

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
400	590	35	-

Materials to be welded

Ti-, Nb Stabilized	DIN 17 440/17 445	X6 CrNiTi 18 10	1.4541
		X6 CrNiNb 18 10	1.4550
		G-X5 CrNiNb 18 9	1.4552
Non Stabilized	DIN 17 440/17 445	X5 CrNi 18 10	1.4301
		X2 CrNi 19 11	1.4306
		G-X6 CrNi 18 9	1.4308
		G-10 CrNi 18 8	1.4312

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
5/64	2.0	12	300	35-55	1.1	1.5/2.0	9.0/12.0
3/32	2.5	12	300	50-80	1.7	1.5/2.0	9.0/12.0
1/8	3.25	14	350	80-110	3.3	2.0/2.5	12.0/15.0
5/32	4.0	14	350	120-150	4.9	2.0/2.5	12.0/15.0
3/16	5.0	14	350	150-180	7.1	2.0/2.5	12.0/15.0

Re-drying :300^{0}C/2h

Stainless Steel Electrodes

Z-ARM BS

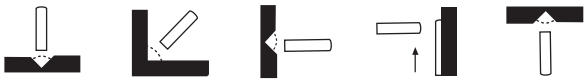
Description and Applications:

- *Basic coated austenitic stainless steel electrode especially designed for welding armour plate and other difficult to weld steels*
- *All position welding on DC current*
- *Neat weld appearance*
- *Excelent slag release*

Specifications:

AWS/ASME SFA A5.4 : E 308Mo-15 mod
BS 2926 : E 19 9 3 B

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni	Mo
0.05	1.90	0.30	20.50	9.80	2.90

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
540	730	25	-

Materials to be welded

- Forged and cast armour steels, alloyed steels, high carbon steels, Mn steel, high strength steels, spring steels and stainless steels
- Joining dissimilar steels and steels having different chemical composition

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
1/8	3.25	14	350	90-120	3.2	2.0/2.5	12.0/15.0
5/32	4.0	14	350	135-185	4.7	2.0/2.5	12.0/15.0
3/16	5.0	14	350	175-255	6.9	5.0/6.0	12.0/18.0

Re-drying : 300°C/2h

Stainless Steel Electrodes

Z-Ni Vi

Description and Applications:

- *Rutile coated austenitic stainless steel electrode especially designed for welding armour plates and other difficult to weld steels*
- *All position welding on both AC and DC*
- *Neat weld appearance*
- *Excellent slag release*

Specifications:

AWS/ASME SFA A5.4 : E 308Mo-16 mod
BS 2926 : E 19 9 3 R

Welding position

ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 70V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni	Mo
0.03	0.85	0.75	19.50	9.80	2.60

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
500	680	35	90 J @ 20°C

Materials to be welded

- Forged and cast armour steels, alloyed steels, high carbon steels, Mn steel, high strength steels, spring steels and stainless steels
- Joining dissimilar steels and steels having different chemical composition

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
5/64	2.0	12	300	35-60	1.2	1.5/2.0	9.0/12.0
3/32	2.5	12	300	50-80	1.8	1.5/2.0	9.0/12.0
1/8	3.25	14	350	90-120	3.3	2.0/2.5	12.0/15.0
5/32	4.0	14	350	130-170	5.0	2.0/2.5	12.0/15.0
3/16	5.0	14	350	160-220	7.5	2.0/2.5	12.0/15.0

Re-drying :300°C/2h

Stainless Steel Electrodes

Z-2209 B

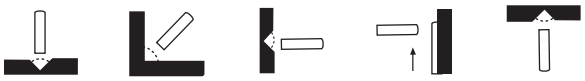
Description and Applications:

- *Basic coated electrode for welding stainless ferritic-austenitic Cr-Ni-Mo Steels (Duplex steels)*
- *Highly resistant to pitting and stess corrosion craking*
- *Recommended for sub-zero toughness, positional welding applications*
- *Maximum temperature of use +250^{0}C*

Specifications:

AWS/ASME SFA A5.4 : E 2209-15
DIN 8556: E 22 9 3 L B 20+
EN 1600 : E 22 9 3 L B 22

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni	Mo	N
<0.04	1.00	0.40	25.50	9.00	3.00	0.14

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
620	770	25	100 J @ 20°C

Materials to be welded

Duplex Stainless-Steels	DIN/SEW 400	X2 CrNiMoN 22 5 3	1.4462

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	50-70	1.6	1.5	9.0
1/8	3.25	18	450	70-100	3.5	2.0	12.0
5/32	4.0	18	450	100-150	5.3	2.0	12.0

Re-drying :300^{0}C/2h

Stainless Steel Electrodes

Z-2209 R

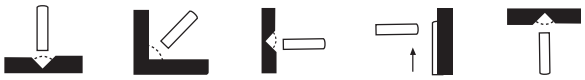
Description and Applications:

- *Rutile-basic coated electrode for welding stainless ferritic-austenitic Cr-Ni-Mo Steels (Duplex steels)*
- *Highly resistant to pitting and stress corrosion craking*
- *Suited for all-position welding applications on both AC and DC*
- *Maximum temperature of use +250°C*

Specifications:

AWS/ASME SFA A5.4 : E 2209-16
DIN 8556: E 22 9 3 L R 23
EN 1600 : E 22 9 3 L R 12

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 50V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni	Mo	N
0.03	0.90	0.80	25.50	9.00	3.00	0.14

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
620	750	25	60 J @ 20°C

Materials to be welded

Duplex Stainless-Steels	DIN/SEW 400	X2 CrNiMoN 22 5 3	1.4462

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	60-70	1.6	1.5	9.0
1/8	3.25	18	450	70-100	3.5	2.0	12.0
5/32	4.0	18	450	90-140	5.3	2.0	12.0

Re-drying :300°C/2h

Stainless Steel Electrodes

Z-2594 R

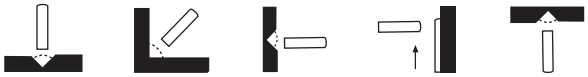
Description and Applications:

- *Rutile electrode designed for welding ferritic- austenitic steels of Super Duplex types.*
- *Its chemical composition allows high resistance to pitting and intergranular corrosion combined to excellent mechanical properties.*
- *Well suited for condition in the off-shore field.*

Specifications:

AWS/ASME SFA 5.4 : E 2594-16
EN 1600 2594 NLR 2 3

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 50V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni	Mo	Cu	N	FN
0.03	0.9	0.9	25.25	9.8	4.0	0.75	0.25	30-50

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A5 [%]	Impact [ISO-V]
700	850	26	60 J@20°C 40J@-40°C

Materials to be welded

Super Duplex Stainless-Steel

		ASTM	EN
SAF	2507	S32750	1.4410
ZERON100		S32760	

1.4462

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	50-75	2	12
1/8	3.25	14	450	70-120	2	12
5/32	4.0	14	450	90-160	2	12

Hard Facing Electrodes

Z-104

Description and Applications:

- *Basic coated abrasion resistant electrode*
- *Machinable weld deposit*
- *Moderate impact toughness*
- *Applicable for gear teeth, brake drums, sprockets, bushings and rail crossings*

Specifications:

DIN 8555: E 1 UM 300

Welding position

ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 70V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Fe
0.10	0.50	0.80	3.40	Bal.

Typical all weld metal mechanical properties

Hardness	
28-33 Rc	275-325 HB

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	60-80	2.1	5.0	15.0
1/8	3.25	18	350/450	110-130	4.5	5.0/6.0	15.0/18.0
5/32	4.0	18	450	140-180	6.7	6.0	18.0
3/16	5.0	18	450	190-240	11.2	6.0	18.0

Re-drying :300^{0}C/2h

Hard Facing Electrodes

Z-105

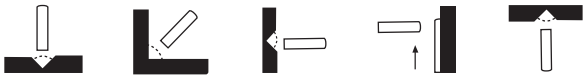
Description and Applications:

- *Basic coated electrode for surfacing parts subjected to medium abrasion and high compression*
- *Applicable for runners, caterpillar track links, gear wheels and dredging machinery*

Specifications:

DIN 8555: E 1 UM 400

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 70V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Fe
0.30	0.30	0.70	3.20	Bal.

Typical all weld metal mechanical properties

	Hardness
37-42Rc	350-400 HB

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	70-100	2.2	5.0	15.0
1/8	3.25	14/18	350/450	105-115	3.5/4.5	5.0/6.0	15.0/18.0
5/32	4.0	18	450	120-160	6.6	6.0	18.0
3/16	5.0	18	450	180-240	10.3	6.0	18.0
1/4	6.0	18	450	250-290	15.0	6.0	18.0

Re-drying :300°C/2h

Hard Facing Electrodes

Z-110

Description and Applications:

- *Rutile coated electrode for corrosion resistant hard surfacing up to 500°C*
- *Applicable for shafts, rails, mixer arms, valve seats of cast steel leading buckets*

Specifications:

DIN 8555: E 6 UM 55 R

Welding position

ASME:	1G	2F	2G	3G
ISO 6947	PA	PB	PC	PF

Current

= +	~ (Uo : 70V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Mo	Fe
0.30	0.40	0.70	11.90	0.90	Bal.

Typical all weld metal mechanical properties

Hardness	
53-56 Rc	525-580 HB

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
1/8	3.25	14	350	85-120	3.0	5.0	15.0
5/32	4.0	14/18	350	120-150	4.5/5.8	5.0	15.0
3/16	5.0	18	450	160-210	9.1	6.0	18.0

Re-drying :120°C/1h

Hard Facing Electrodes

Z-111

Description and Applications:

- *Rutile coated electrode for hard surfacing of earth moving machines*
- *Low open current voltage*
- *Heat resistant up to 600°C*
- *Applicable for buldozer blades, crusher jaws, excavator teeth and mixers*
  Non machinable weld deposit

Specifications:

DIN 8555: E 6 UM 55

Welding position

ASME:	1G	2F	2G	3G
ISO 6947	PA	PB	PC	PF

Current

= +	~ (Uo : 50V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Mo	Fe
0.50	1.10	0.40	7.20	0.90	Bal.

Typical all weld metal mechanical properties

Hardness	
55-58Rc	560-615 HB

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	70-100	2.1	5.0	15.0
1/8	3.25	14/18	350/450	90-110	3.5/4.6	5.0/6.0	15.0/18.0
5/32	4.0	14/18	350/450	140-180	5.3/6.8	5.0/6.0	15.0/18.0
3/16	5.0	18	450	180-250	10.6	6.0	18.0
1/4	6.0	18	450	220-280	15.0	6.0	18.0

Re-drying : 120°C/1h

Hard Facing Electrodes

Z-113

Description and Applications:

- *Basic coated electrode for surfacing parts subjected to impact, high compression stresses and severe abrasion*
- *Applicable for tracctor and crane parts, shovel teeth, crusher hammers and scraper blades*

Specifications:

DIN 8555: E 6 UM 60

Welding position

ASME:	1G	2F	2G	3G
ISO 6947	PA	PB	PC	PF

Current

= +	~ (Uo : 70V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Mo	V	Fe
0.50	0.40	0.80	7.50	0.50	0.60	Bal.

Typical all weld metal mechanical properties

Hardness
57-62 Rc

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	60-80	2.1	5.0	15.0
1/8	3.25	14/18	350/450	100-130	3.5/4.4	5.0/6.0	15.0/18.0
5/32	4.0	14/18	350/450	120-160	5.3/6.7	5.0/6.0	15.0/18.0
3/16	5.0	18	450	180-210	10.6	6.0	18.0
1/4	6.0	18	450	220-280	15.0	6.0	18.0

Re-drying : 300^{0}C/2h

Hard Facing Electrodes

Z-114

Description and Applications:

- *Rutile coated electrode for surfacing parts subjected to abrasion and impact*
- *Applicable for rails truck wheels, worms, coupling shafts and spindles*
- *Non machinable weld deposit*

Specifications:

DIN 8555: E 6 UM 60

Welding position

ASME:	1G	2F	2G	3G
ISO 6947	PA	PB	PC	PF

Current

= +	~ (Uo : 70V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Mo
0.60	0.90	0.70	4.70	1.80

Typical all weld metal mechanical properties

	Hardness
58-60 Rc	620-650 HB

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	75-110	2.4	5.0	15.0
1/8	3.25	14	350	110-140	3.9	5.0	15.0
5/32	4.0	14/18	350/450	150-190	6.1/7.8	5.0/6.0	15.0/18.0
3/16	5.0	18	450	190-260	12.3	6.0	18.0

Re-drying : 120^{0}C/1h

Hard Facing Electrodes

Z-120

Description and Applications:

- *Rutile coated electrode for high abrasion without impact applications*
- *160% weld metal recovery*
- *Can be used both for buildups and rebuilding*
- *Suitable for crushing mills, rock drills, mixers and plough shares*

Specifications:

DIN 8555: E 10 UM 60 Z

Welding position

ASME:	1G	2F
ISO 6947	PA	PB

Current

= +	~ (Uo : 70V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Fe
3.00	0.20	0.90	29.00	Bal.

Typical all weld metal mechanical properties

	Hardness
60 Rc	60-650 HB

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
1/8	3.25	14	350	100-130	5.6	4.0	12.0
5/32	4.0	14	350	120-160	8.3	4.0	12.0
3/16	5.0	14	350	150-240	13.3	4.0	12.0

Re-drying : 300°C/1-2h

Hard Facing Electrodes

Z-120 B

Description and Applications:

- *Basic coated electrode for surfacing parts subjected to severe abrasion*
- *Applicable for crusher parts, conveyer screws and bucket teeth*

Specifications:

DIN 8555: E 10 UM 60 GR

Welding position

ASME:	1G	2F
ISO 6947	PA	PB

Current

= +

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Fe
3.90	0.30	1.00	24.90	Bal.

Typical all weld metal mechanical properties

Hardness	
60 Rc	650 HB

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
1/8	3.25	14	350	110-140	5.3	4.5	13.5
5/32	4.0	18	450	140-180	10.6	6.0	18.0
3/16	5.0	18	450	185-230	16.3	6.0	18.0

Re-drying : 300^{0}C/1-2h

Hard Facing Electrodes

Z-123

Description and Applications:

- *Basic coated electrode for hardsurfacing parts subjected to severe abrasion without impact*
- *Use up to 450^{0}C*
- *Applicable for screws of refractory material presses, brick presses and cement presses*

Specifications:

DIN 8555: E 10 UM 65 GR

Welding position

ASME:	1G	2F
ISO 6947	PA	PB

Current

= +	~	(Uo : 70V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Nb	Fe
6.50	0.80	0.60	22.70	7.00	Bal.

Typical all weld metal mechanical properties

Hardness	
64 Rc	720 HB

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
1/8	3.25	14	350	115-150	5.7	4.5	13.5
5/32	4.0	18	450	135-200	11.5	6.0	18.0
3/16	5.0	18	450	195-255	17.6	6.0	18.0

Re-drying : 300^{0}C/1-2h

Hard Facing Electrodes

Z-125

Description and Applications:

- *Basic coated electrode for hardsurfacing parts subjected to severe abrasion without impact*
- *Use up to 650°C*
- *Applicable for sinter crushers, scraper wires in zinc industries, scrap collectors and mechanical stokers*

Specifications:

DIN 8555: E 10 UM 65 GR

Welding position

ASME:	1G	2F
ISO 6947	PA	PB

Current

= +

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Mo	Nb	V	W	Fe
4.30	0.70	0.80	17.70	4.90	2.00	0.60	3.00	Bal.

Typical all weld metal mechanical properties

Hardness	
66 Rc	760 HB

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
1/8	3.25	14	350	125-160	6.3	4.5	13.5
5/32	4.0	18	450	150-200	12.5	6.0	18.0
3/16	5.0	18	450	220-300	19.7	6.0	18.0

Re-drying : 300°C/1-2h

Hard Facing Electrodes

Z-130

Description and Applications:

- *Basic coated electrode for high speed tool surfacing*
- *Moderate resistance to impact*
- *Heat resistant up to 600^{0}C*
- *Applicable for cutting tools, drills, knives, shears, punches and dies*

Specifications:

AWS/ASME SFA A5. 13 : EFe5-B mod
DIN 8555 : E 4 UM 60 S

Welding position

ASME:	1G	2F	2G
ISO 6947	PA	PB	PC

Current

= +

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Mo	W	V	Fe
0.70	0.70	0.80	3.70	8.00	2.70	0.60	Bal.

Typical all weld metal mechanical properties

Hardness	
60-65 Rc	650-740 HB

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	80-100	2.7	4.5	13.5
1/8	3.25	14	350	110-130	4.3	4.5/5.0	13.5/15.0
5/32	4.0	14	350	130-160	6.7	4.5/5.0	13.5/15.0

Re-drying : 300^{0}C/1-2h

Hard Facing Electrodes

Z-151

Description and Applications:

- *Basic coated cold work hardening Mn austenitic electrode*
- *High resistance to impact and shock*
- *Used for rebuilding and joining between Mn steel and Carbon steel*
- *Applicable for crusher plates, dredger teeth and hammers*

Specifications:

AWS/ASME SFA A5. 13 : EFeMn-A
DIN 8555: E 7 UM 250 KP

Welding position

ASME:	1G	2F	2G	3G
ISO 6947	PA	PB	PC	PF

Current

= +

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni	Fe
0.60	12.20	0.60	0.10	4.60	Bal.

Typical all weld metal mechanical properties

Hardness

As welded	After work hardening
230 HB	500 HB

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
1/8	3.25	14/18	350/450	100-130	3.7	5.0/6.0	15.0/18.0
5/32	4.0	14/18	350/450	150-180	6.0	5.0/6.0	15.0/18.0
3/16	5.0	18	450	200-250	11.1	6.0	18.0
1/4	6.0	18	450	240-290	16.2	6.0	18.0

Re-drying : 300°C/1-2h

Hard Facing Electrodes

Z-151 NC

Description and Applications:

- *Basic coated cold work hardening Mn austenitic electrode*
- *High resistance to impact and shock*
- *Used for rebuilding and joining between Mn steel and Carbon steel*
- *Applicable for crusher plates, dredger teeth and hammers*
- *Non conductive coating*

Specifications:

AWS/ASME SFA A5. 13 : EFeMn-A mod
DIN 8555: E 7 UM 250 KP

Welding position

ASME:	1G	2F	2G	3G
ISO 6947	PA	PB	PC	PF

Current

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Ni	Fe
1.00	13.20	0.25	4.60	5.00	Bal.

Typical all weld metal mechanical properties

Hardness

As welded	After work hardening
200 HB	520 HB

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
1/8	3.25	14	350	95-130	3.9	5.0	15.0
5/32	4.0	14	350	130-180	6.3	5.0	15.0
3/16	5.0	18	450	190-250	11.2	6.0	18.0
1/4	6.0	18	450	240-280	17.0	6.0	18.0

Re-drying : 300^{0}C/1-2h

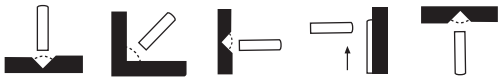
Hard Facing Electrodes

Z-152

Description and Applications:

- *Rutile coated electrode for surfacing parts subjected to high impact and cavitation*
- *Joining and surfacing Mn steels and hard to weld steels*
- *Applicable for dredge pumps, hydraulic press pistons, crushing equipment and rails*

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 70V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Fe
0.60	12.90	0.30	14.70	Bal.

Typical all weld metal mechanical properties

Hardness

As welded	After work hardening
200 HB	500 HB

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
1/8	3.25	18	450	110-140	6.4	6.0	18.0
5/32	4.0	18	450	165-175	9.6	6.0	18.0
3/16	5.0	18	450	190-220	13.5	6.0	18.0
1/4	6.0	18	450	260-290	19.5	6.0	18.0

Re-drying : 300^{0}C/1-2h

Hard Facing Electrodes

Z-STEL 6

Description and Applications:

- *Cobalt base hard facing electrode*
- *Used for high temperature high impact applications (up to 900°C)*
- *Corrosion resistant in organic and anorganic media*

Specifications:

AWS/ASME SFA A5. 13 : ECo Cr-A
DIN 8555 : E 20 UM 45 CTZ

Welding position

ASME:	1G	2F
ISO 6947	PA	PB

Current

= +

Typical all weld metal composition [wt%]

C	Cr	W	Co
1.10	28.00	4.50	Bal.

Typical all weld metal mechanical properties

Hardness
44 Rc

Welding parameters and packaging data

Dia		Length		Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
in	mm	in	mm				
1/8	3.25	14	350	90-120	3.7	2.0/2.5	12.0/15.0
5/32	4.0	18	350	120-155	5.9	2.0/2.5	12.0/15.0
3/16	5.0	18	350	150-190	8.5	2.0/2.5	12.0/15.0

Re-drying : 300°C/1-2h

Hard Facing Electrodes

Z-STEL 12

Description and Applications:

- *Cobalt base hard facing electrode*
- *Used for high temperature moderate impact applications (up to 900^{0}C)*
- *Corrosion resistant in organic and anorganic media*

Specifications:

AWS/ASME SFA A5. 13 : ECo Cr-B
DIN 8555 : E 20 UM 50 CTZ

Welding position

ASME:	1G	2F
ISO 6947	PA	PB

Current

= +

Typical all weld metal composition [wt%]

C	Cr	W	Co
1.40	29.00	8.00	Bal.

Typical all weld metal mechanical properties

Hardness
47 Rc

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
1/8	3.25	14	350	90-120	3.9	2.0/2.5	12.0
5/32	4.0	18	350	120-155	5.9	2.0/2.5	12.0
3/16	5.0	18	350	150-190	7.3	2.0/2.5	12.0

Re-drying : 300^{0}C/1-2h

Non Ferrous Electrodes

Z-Ni 1

Description and Applications:

- *Basic coated pure nickel electrode*
- *Especially for welding nickel and nickel alloys and joining to unalloyed and low alloyed steels*
- *Suitable for welding NiCu and NiCu alloys to carbon, stainless and nickel base alloys*

Specifications:

AWS/ASME SFA A5. 11 : ENi-1
DIN 1736: EL-NiTi 3

Welding position

ASME:	1G	2F	2G	3G
ISO 6947	PA	PB	PC	PF

Current

Typical all weld metal composition [wt%]

C	Mn	Si	Ti	Fe	Ni
0.025	0.40	0.50	2.30	0.30	Bal.

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
310	470	30	-

Materials to be welded

ASTM	UNS	DIN	W.Nr
Alloy 200	N02200	Ni99.2	2.4066
Alloy 201	N02201	LC Ni99	2.4068
Alloy 205	N02205	LC Ni99.6	2.4061
-	-	Ni99.6	2.4060
-	-	Ni99.8	2.4050
-	-	Ni99.4Fe	2.4062
-	-	NiMn1	2.4106
-	-	NiMn2	1.4210
-	-	NiMn3Al	2.4122
-	-	NiMn5	2.4116
-	-	NiA14Ti	2.4128

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	50-80	2.0	2.0/2.5	12.0/15.0
1/8	3.25	14	350	75-120	3.3	2.0/2.5	12.0/15.0
5/32	4.0	14	350	100-140	5.1	2.0/2.5	12.0

Re-drying :250^{0}C/1-2h

Non Ferrous Electrodes

Z-Ni B

Description and Applications:

- *Basic coated thick electrode for depositing NiMo-1 alloy and welding NiMo alloys*
- *Can be used at very high temperatures in reducing gases environment*
- *Excellent resistance to acetic, phosphoric, sulphuric and hydrochloric acids*

Specifications:

AWS/ASME SFA A5. 11 : ENiMo-1
DIN 1736 : EL-NiMo 29

Welding position

ASME:	1G	2F	2G	3G
ISO 6947	PA	PB	PC	PF

Current

= +	~ (Uo : 70V)

Typical all weld metal composition [wt%]

C	Mn	Si	Fe	Mo	V	Ni
0.05	0.60	0.60	5.00	27.00	0.25	Bal.

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
500	700	26	-

Materials to be welded

ASTM	UNS	DIN	W.Nr
Alloy B2	N10665	NiMo29	2.4616

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	90-110	3.6	2.0/2.5	12.0/15.0
1/8	3.25	14	350	130-150	6.3	2.0/2.5	12.0/15.0
5/32	4.0	14	350	180-200	9.3	2.0/2.5	12.0/15.0

Re-drying :250°C/2h

Non Ferrous Electrodes

Z-Ni A2

Description and Applications:

Basic coated electrode designed for joining Nickel base alloys and for joining dissimilar steels. The welding designed to endure wide range of temp from cryogenic up to 980°C

Specifications:

AWS/ASME SFA A5. 11 : ENiCrFe-2
DIN 1736: EL-NiCr 15 FeNb

ASME:	1G	2F	2G	3G
ISO 6947	PA	PB	PC	PF

Current

= +	~ (Uo : 70V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Mo	Nb	Fe	Ni
0.07	3.40	0.40	16.10	1.80	2.10	6.60	Bal.

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
410	650	32	50 J @ -196°C

Materials to be welded

ASTM	UNS	DIN	W.Nr
Alloy 800	N08800	X10 NiCrAlTi32 20	1.4876
Alloy 600	N06600	NiCr 15Fe	2.4640
Alloy B	N06004	NiCr60 15	2.4867
Alloy C	N06003	NiCr80 20	2.4869
-	-	NiCr10	2.4870
Alloy 75	N06075	NiCr20Ti	2.4951
Alloy 80A	N07080	NiCr20TiAl	2.4952
Alloy 9%Ni	K81340	X8 Ni9	1.5662

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	70-90	2.6	2.0/2.5	12.0/15.0
1/8	3.25	14	350	100-135	4.3	2.0/2.5	12.0/15.0
5/32	4.0	14	350	130-180	6.5	2.0/2.5	12.0/15.0
3/16	5.0	14	350	190-220	10.2	2.0/2.5	12.0/15.0

Re-drying : 300°C/1-2h

Non Ferrous Electrodes

Z-NI 182

Description and Applications:

- *Synthetic basic coated NiCr electrode*
- *For welding Ni-base alloys, claddings and dissimilar metals*
- *High creep resistant and high resistance to embrittlement*

Specifications:

AWS/ASME SFA A5. 11 : ENiCrFe-3
DIN 1736: EL-NiCr 15 FeMn

Welding position

ASME:	1G	2F	2G	3G
ISO 6947	PA	PB	PC	PF

Current

= +

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Nb	Fe	Ni
0.02	7.50	0.40	13.80	1.80	4.40	Bal.

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
380	620	32	80 J @ -196°C

Materials to be welded

ASTM	UNS	DIN	W.Nr
Alloy 600	N06600	NiCr15Fe	2.4640
Alloy 601	N06601	NiCr23Fe	2.4851
Alloy 800	N08800	X10 NiCrAlTi32 20	1.4876
Alloy 800H	N08810	X10 NiCrAlTi32 20	1.4858
Alloy 75	N06075	NiCr20Ti	2.4951
Alloy 80A	N07080	NiCr20TiAl	2.4952
Alloy 90	N07090	NiCr20Co18Ti	2.4969
Alloy 9%Ni	K81340	X8 Ni9	1.5662

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	75-100	2.6	2.0/2.5	12.0/15.0
1/8	3.25	14	350	100-140	4.6	2.0/2.5	12.0/15.0
5/32	4.0	14	350	140-180	7.0	2.0/2.5	12.0/15.0
3/16	5.0	14	350	170-210	10.0	2.0/2.5	12.0/15.0

Re-drying : 250°C/1-2h

Non Ferrous Electrodes

Z-NI 182R

Description and Applications:

- *Rutile basic coated NiCr electrode*
- *For welding Ni-base alloys, claddings and dissimilar metals*
- *High creep resistant and high resistance to embrittlement*

Specifications:

AWS/ASME SFA A5. 11 : ENiCrFe-3
DIN 1736: EL-NiCr 15 FeMn

Welding position

ASME:	1G	2F	2G	3G
ISO 6947	PA	PB	PC	PF

Current

= +	~ (Uo : 70V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Nb	Fe	Ni
0.02	6.70	0.70	13.50	1.60	5.00	Bal.

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
400	580	32	110 J @ -196°C

Materials to be welded

ASTM	UNS	DIN	W.Nr
Alloy 600	N06600	NiCr15Fe	2.4640
Alloy 601	N06601	NiCr23Fe	2.4851
Alloy 800	N08800	X10 NiCrAlTi32 20	1.4876
Alloy 800H	N08810	X10 NiCrAlTi32 20	1.4858
Alloy 75	N06075	NiCr20Ti	2.4951
Alloy 80A	N07080	NiCr20TiAl	2.4952
Alloy 90	N07090	NiCr20Co18Ti	2.4969
Alloy 9%Ni	K81340	X8 Ni9	1.5662

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	105-145	2.8	2.0/2.5	12.0/15.0
1/8	3.25	14	350	140-180	4.8	2.0/2.5	12.0/15.0
5/32	4.0	14	350	175-215	7.3	2.0/2.5	12.0/15.0
3/16	5.0	14	350	225-265	11.2	2.0/2.5	12.0/15.0

Re-drying : 250°C/1-2h

Non Ferrous Electrodes

Z-Ni 112

Description and Applications:

- *Synthetic basic coated NiCrMo electrode*
- *Suitable for welding alloy 625, alloy 825 or similar grades*
- *Joining 9%Ni for high toughness at cryogenic temperatures*
- *Extreme high rsistance to general and intergranular corrosion, pitting, crevice and stess corrosion cracking*
- *Resistant to high temperature oxidaton and carburization*

Specifications:

AWS/ASME SFA A5. 11 : ENiCrMo-3
DIN 1736: EL-NiCr 20 Mo 9 Nb

Welding position

ASME:	1G	2F	2G	3G
ISO 6947	PA	PB	PC	PF

Current

= +	~ (Uo : 70V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Nb	Mo	Fe	Ni
0.05	0.90	0.60	21.0	3.4	9.5	3.4	Bal.

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
530	780	38	70 J @ -196°C

Materials to be welded

ASTM	UNS	DIN	W.Nr
Alloy 625	N06625	NiCr22Mo9Nb	2.4856
Alloy 825	N08825	NiCr21Mo	2.4858
Alloy 600	N06600	NiCr15Fe	2.4640
Alloy 800	N08800	X10 NiCrAlTi32 20	1.4876
Alloy 75	N06075	NiCr20Ti	2.4951
Alloy 80A	N07080	NiCr20TiAl	2.4952
Alloy 9%Ni	K81340	X8 Ni9	1.5662
Alloy 6%Mo	S31254		

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	90-130	2.7	2.0/2.5	12.0/15.0
1/8	3.25	14	350	120-160	4.6	2.0/2.5	12.0/15.0
5/32	4.0	14	350	150-190	7.0	2.0/2.5	12.0/15.0
3/16	5.0	14	350	190-230	10.7	2.0/2.5	12.0/15.0

Re-drying : 250°C/1h

Non Ferrous Electrodes

Z-Ni 112R

Description and Applications:

- *Synthetic rutile coated NiCrMo electrode*
- *Suitable for welding alloy 625, alloy 825 or similar grades*
- *Joining 9%Ni for high toughness at cryogenic temperatures*
- *Extreme high rsistance to general and intergranular corrosion, pitting, crevice and stess corrosion cracking*
- *Resistant to high temperature oxidaton and carburization*

Specifications:

AWS/ASME SFA A5. 11 : ENiCrMo-3
DIN 1736: EL-NiCr 20 Mo 9 Nb

Welding position

ASME:	1G	2F	2G	3G
ISO 6947	PA	PB	PC	PF

Current

= +	~ (Uo : 70V)

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Nb	Mo	Fe	Ni
0.04	0.70	0.70	20.00	3.30	9.50	3.50	Bal.

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
420	760	32	70 J @ -196°C

Materials to be welded

ASTM	UNS	DIN	W.Nr
Alloy 625	N06625	NiCr22Mo9Nb	2.4856
Alloy 825	N08825	NiCr21Mo	2.4858
Alloy 600	N06600	NiCr15Fe	2.4640
Alloy 800	N08800	X10 NiCrAlTi32 20	1.4876
Alloy 75	N06075	NiCr20Ti	2.4951
Alloy 80A	N07080	NiCr20TiAl	2.4952
Alloy 9%Ni	K81340	X8 Ni9	1.5662
Alloy 6%Mo	S31254		

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	90-130	4.4	2.0/2.5	12.0/15.0
1/8	3.25	14	350	120-160	5.9	2.0/2.5	12.0/15.0
5/32	4.0	14	350	150-190	7.3	2.0/2.5	12.0/15.0
3/16	5.0	14	350	190-230	8.8	2.0/2.5	12.0/15.0

Re-drying : 250°C/1h

Non Ferrous Electrodes

Z-NiCroMo3

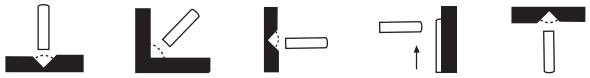
Description and Applications:

- *All position electrode depositing austenitic Ni-base high CrMoNb weld*
- *Matching core wire*
- *Especially designed to match th composition ans properties of alloy 625*
- *Extremely high resistance to general and intergranular corrosion, pitting and crevice corrosion and strees corrosion cracking*
- *Resistant to stress corrosion cracking in sulphuric acid solutions and chloride containing environments*
- *Resistant to hot cracking and high temperature oxidation and carburiztion*
- *High toughness at cryogenic temperatures*

Specifications:

AWS/ASME SFA A5. 11 : ENiCrMo-3
DIN 1736: EL-NiCr 20 Mo 9 Nb

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Mo	Nb	Fe	Ni
0.03	0.40	0.50	21.50	9.50	3.50	3.00	Bal.

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
470	780	37	40 J @ -196°C

Materials to be welded

ASTM	UNS	DIN	W.Nr
Alloy 625	N06625	NiCr22Mo9Nb	2.4856
Alloy 825	N08825	NiCr21Mo	2.4858
Alloy 600	N06600	NiCr15Fe	2.4640
Alloy 800	N08800	X10 NiCrAlTi32 20	1.4876
Alloy 75	N06075	NiCr20Ti	2.4951
Alloy 80A	N07080	NiCr20TiAl	2.4952
Alloy 9%Ni	K81340	X8 Ni9	1.5662
Alloy 6%Mo	S31254		

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	10	250	40-70	1.9	2.0	12.0
1/8	3.25	12	300	70-100	2.9	2.0	12.0
5/32	4.0	12	300	90-140	3.7	2.0	12.0

Re-drying : 250°C/1-2h

Non Ferrous Electrodes

Z-NiCroMo4

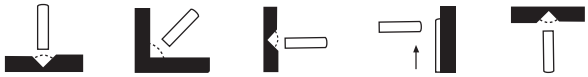
Description and Applications:

- *All position electrode depositing austenitic Ni-base high CrMoNb weld*
- *Matching core wire*
- *Especially designed to match th composition ans properties of alloy C276*
- *Extremely high resistance to general and intergranular corrosion, pitting and crevice corrosion and strees corrosion cracking*
- *Resistant to stress corrosion cracking in sulphuric acid solutions and chloride containing environments*
- *Resistant to hot cracking and high temperature oxidation and carburiztion*
- *High toughness at cryogenic temperatures*

Specifications:

AWS/ASME SFA A5. 11 : ENiCrMo-4
DIN 1736: EL-NiMo 15 Cr 15 W

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Mo	W	Fe	Ni
0.015	0.30	0.10	16.00	16.50	3.50	6.00	Bal.

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
560	770	30	50 J @ -196°C

Materials to be welded

ASTM	UNS	DIN	W.Nr
Alloy C276	N10276	NiMo16Cr15W	2.4819
Alloy C4	N06455	NiMo16Cr15Ti	2.4610
Alloy 625	N06625	NiCr22Mo9Nb	2.4856
Alloy G	N06007	NiCr22Mo6Cu	2.4618
Alloy G3	N06985	NiCr22Mo7Cu	2.4619

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	10	250	40-70	1.9	2.0	12.0
1/8	3.25	12	300	70-100	2.9	2.0	12.0
5/32	4.0	12	300	90-140	3.7	2.0	12.0

Re-drying : 250°C/1-2h

Non Ferrous Electrodes

Z-Ni C

Description and Applications:

- *Basic coated electrode giving a Ni-base weld metal equivalent in composition of Hastelloy C*
- *Ideal for use as build ups in high temperatures oxidation, corrosion and mechanical stress applications*

Specifications:

AWS/ASME SFA A5. 11 : ENiCrMo-5
DIN 1736: EL-NiMo 15 Cr 15 W mod

Welding position

ASME:	1G	2F	2G	3G
ISO 6947	PA	PB	PC	PF

Current

Typical all weld metal composition [wt%]

C	Mn	Si	Cr	Mo	Fe	W	Ni
0.04	0.30	0.90	14.50	16.80	4.00	3.70	Bal.

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
-	720	28	-

Materials to be welded

ASTM	UNS	DIN	W.Nr
Alloy C276	N10276	Ni Mo 16Cr 15W	2.4819
Alloy C4	N06455	Ni Mo 16Cr 15Ti	2.4610
Alloy C625	N06625	Ni Cr 22Mo 9Nb	2.4856
		Ni Cr 22Mo 6Cu	2.4618
		Ni Cr 22Mo 7Cu	2.4619

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	70-90	3.6	2.0/2.5	12.0/15.0
1/8	3.25	14	350	100-110	6.3	2.0/2.5	12.0/15.0
5/32	4.0	14	350	130-160	9.3	2.0/2.5	12.0/15.0
3/16	5.0	14	350	160-200	14.5	2.0/2.5	12.0/15.0

Re-drying : 250°C/1-2h

Non Ferrous Electrodes

Z-Ni 190

Description and Applications:

- *Basic coated Ni-Cu electrode for welding Ni-Cu alloy to mild and low alloy steels*
- *High resistance to sea water corrosion*
- *Excellent resistance to stress corrosion cracking*

Specifications:

AWS/ASME SFA A5. 11 : ENiCu-7
DIN 1736: EL-NiCu 30 Mn

Welding position

ASME:	1G	2F	2G	3G
ISO 6947	PA	PB	PC	PF

Current

Typical all weld metal composition [wt%]

C	Mn	Si	Cu	Ti	Fe	Ni
0.03	2.60	0.60	33.00	0.30	1.30	Bal.

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]
320	510	35	120 J @ -20°C

Materials to be welded

ASTM	UNS
Monel 400	N04400
Monel K500	N65500

DIN 17 743	W.Nr
NiCu 30 Fe	2.4360
LC NiCu 30 Fe	2.4361
G Ni Cu 30 Nb	2.4365
NiCu30 Al	2.4375
NiCu 14 FeMo	2.4400

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	70-90	1.9	2.0/2.5	12.0/15.0
1/8	3.25	14	350	90-125	3.2	2.0/2.5	12.0/15.0
5/32	4.0	14	350	110-160	4.9	2.0/2.5	12.0/15.0
3/16	5.0	14	350	155-185	7.5	2.0/2.5	12.0/15.0

Re-drying : 300°C/2h

Non Ferrous Electrodes

Z-Ni 187

Description and Applications:

- *Basic coated electrode for welding wrought iron or cast copper nickel alloys (max 30% Ni)*
- *Can be used for cladding steels*
- *Weld metal Resistant to Corrosion, erosion and cavitation in sea-water*

Specifications:

AWS/ASME SFA A5.6 : ECuNi
DIN 1733 : EL-CuNi 30 Mn

Welding position

ASME:	1G	2F	2G	4G
ISO 6947	PA	PB	PC	PE

Current

= +	~

Typical all weld metal composition [wt%]

Mn	Si	Ni	Fe	Cu
1.70	0.70	30.00	0.60	Bal.

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Hardness
220	370	30	105 HB

Materials to be welded

UNS	DIN	W.Nr
C71500	CuNi 30Mn1Fe	2.0882
C71300	CuNi 25	2.0830
C70600	CuNi 10Fe	2.0872

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	50-75	2.2	2.0/2.5	12.0/15.0
1/8	3.25	14	350	75-110	3.4	2.0/2.5	12.0/15.0
5/32	4.0	14	350	110-145	5.0	2.0/2.5	12.0/15.0
3/16	5.0	14	350	145-185	7.9	2.0/2.5	12.0/15.0

Re-drying : 300°C/1-2h

Non Ferrous Electrodes

Z-BRONZE A

Description and Applications:

- *Basic coated electrode for joining phosphor-bronze plates*
- *Applicable for electrial conductors*
- *Used for surfacing bronze, cast steel and copper plates*

Specifications:

AWS/ASME SFA A5. 6 : ECuSn-A

Welding position

ASME:	1G	2F
ISO 6947	PA	PB

Current

= +

Typical all weld metal composition [wt%]

Sn	P	Cu
5.00	0.30	Bal.

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	Hardness
190	250	100 HB

Materials to be welded

UNS	DIN	W.Nr
C50700	CuSn2	2.1010
C51100	CuSn4	2.1016

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	40-65	1.9	2.0/2.5	12.0/15.0
1/8	3.25	14	350	70-100	3.1	2.0/2.5	12.0/15.0
5/32	4.0	14	350	120-160	4.8	2.0/2.5	12.0/15.0
3/16	5.0	14	350	160-200	6.8	2.0/2.5	12.0/15.0

Re-drying : 250°C/1-2h

Non Ferrous Electrodes

Z-BRONZE

Description and Applications:

- *Basic coated electrode for assembly and surfacing welds on copper alloys*
- *Ideal for surfacing on steel, cast steel and cast iron*

Specifications:

AWS/ASME SFA A5. 6 : ECuSn-C
DIN 1733 : EL-CuSn 7

Welding position

ASME:	1G	2F
ISO 6947	PA	PB

Current

= +

Typical all weld metal composition [wt%]

Sn	P	Cu
7.80	0.20	Bal.

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	Hardness
140	300	105 HB

Materials to be welded

UNS	DIN	W.Nr
C50700	CuSn2	2.1010
C51100	CuSn4	2.1016
C51900	CuSn6	2.1020
C52100	CuSn8	2.1030

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	40-65	1.9	2.0/2.5	12.0/15.0
1/8	3.25	14	350	70-100	3.1	2.0/2.5	12.0/15.0
5/32	4.0	14	350	120-160	4.8	2.0/2.5	12.0/15.0
3/16	5.0	14	350	160-200	6.8	2.0/2.5	12.0/15.0

Re-drying : 250°C/1-2h

Non Ferrous Electrodes

Z-BRONZE M

Description and Applications:

- *Basic coated electrode for assembly and surfacing welds on copper alloys*
- *Ideal for surfacing on steel, cast steel and cast iron*

Specifications:

AWS/ASME SFA A5. 6 : ECuSn-C
DIN 1733 : EL-CuSn 7

Welding position

ASME:	1G	2F
ISO 6947	PA	PB

Current

= +	~ (Uo : 70V)

Typical all weld metal composition [wt%]

Sn	P	Cu
7.10	0.20	Bal.

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	Hardness
140	300	105 HB

Materials to be welded

UNS	DIN	W.Nr
C50700	CuSn2	2.1010
C51100	CuSn4	2.1016
C51900	CuSn6	2.1020
C52100	CuSn8	2.1030

Welding parameters and packaging data

Dia		Length		Current	Weight	Weight packet	Weight carton
in	mm	in	mm	[amps]	kg/100 pcs	[kg]	[kg]
3/32	2.5	14	350	40-65	1.9	2.0/2.5	12.0/15.0
1/8	3.25	14	350	70-100	3.3	2.0/2.5	12.0/15.0
5/32	4.0	14	350	120-160	5.0	2.0/2.5	12.0/15.0
3/16	5.0	14	350	160-200	7.0	2.0/2.5	12.0/15.0

Re-drying : 250°C/1-2h

Cast Iron & Cutting Electrodes

Z-SuperNi

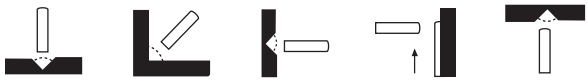
Description and Applications:

- *Special nickel cored electrode for depositing soft, machinable metal on grey cast iron*
- *Suitable for reclamation and repair of grey cast iron and for joining these to mild and C-Mn steels*
- *Can be used with low welding currents*
- *Welding both positive direct current and negative and alternating currents*

Specifications:

AWSASME SFA A5. 15 : ENi-CI
DIN 8573 : E Ni-BG-21

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= ±	~	(Uo : 50V)

Typical all weld metal composition [wt%]

C	Si	Fe	Ni
1.50	3.00	4.00	Bal.

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Hardness
-	300	5	170HB

Materials to be welded

Castings, pump bodies, machine bases, engine blocks, gears, cylinders and valves

Lamellar grey cast iron	DIN 1691	GG 10 to GG 35
Malleable cast iron	DIN 1692	GTS 35 10 to GTS 55 4, GTW 35 04 to GTW 45 07, GTW S 38 12
Nodular cast iron	DIN 1693	GGG 40 to GGG 60

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	50-80	2.0	2.0/2.5	12.0/15.0
1/8	3.25	14	350	80-110	3.3	2.0/2.5	12.0/15.0
5/32	4.0	14	350	110-150	5.1	2.0/2.5	12.0/15.0
3/16	5.0	14	350	150-190	7.4	2.0/2.5	12.0/15.0

Re-drying : 100-150°C/1-2hr

Cast Iron & Cutting Electrodes

Z-Ni 99

Description and Applications:

- *Special electrode for depositing soft machinable metal on grey cast iron*
- *Excellent for repairing casting defects in grey cast iron*
- *Can join cast irons to steels, monels etc.*

Specifications:

AWSASME SFA A5. 15 : ENi-CI
DIN 8573 : E Ni-BG-23

Welding position

ASME:	1G	2F	2G	4G
ISO 6947	PA	PB	PC	PE

Current

= +	~ (Uo : 50V)

Typical all weld metal composition [wt%]

C	Si	Fe	Ni
0.80	0.10	4.50	Bal.

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Hardness
-	300	5	170HB

Materials to be welded

Castings, pump bodies, machine bases, engine blocks, gears, cylinders and valves

Lamellar grey cast iron	DIN 1691	GG 10 to GG 35
Malleable cast iron	DIN 1692	GTS 35 10 to GTS 55 4, GTW 35 04 to GTW 45 07, GTW S 38 12
Nodular cast iron	DIN 1693	GGG 40 to GGG 60

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	50-80	2.0	2.0/2.5	12.0/15.0
1/8	3.25	14	350	80-110	3.4	2.0/2.5	12.0/15.0
5/32	4.0	14	350	110-150	5.2	2.0/2.5	12.0/15.0
3/16	5.0	14	350	150-190	7.4	2.0/2.5	12.0/15.0

Re-drying : 120°C/1-2h

Cast Iron & Cutting Electrodes

Z-Ni 99NC

Description and Applications:

- *Electrode for welding cast iron or malleable cast iron*
- *Non conductive coating*
- *Machinable weld deposit*

Specifications:

AWS/ASME SFA A5. 15 : ENi-CI
DIN 8573 : E Ni-BG-23

Welding position

ASME:	1G	2F	2G	4G
ISO 6947	PA	PB	PC	PE

Current

= +	~ (Uo : 50V)

Typical all weld metal composition [wt%]

C	Si	Fe	Ni
0.75	0.20	5.30	Bal.

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Hardness
-	300	5	170HB

Materials to be welded

Castings, pump bodies, machine bases, engine blocks, gears, cylinders and valves

Lamellar grey cast iron	DIN 1691	GG 10 to GG 35
Malleable cast iron	DIN 1692	GTS 35 10 to GTS 55 4, GTW 35 04 to GTW 45 07, GTW S 38 12
Nodular cast iron	DIN 1693	GGG 40 to GGG 60

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	60-90	2.1	2.0/2.5	12.0/15.0
1/8	3.25	14	350	90-110	3.3	2.0/2.5	12.0/15.0
5/32	4.0	14	350	110-170	5.2	2.0/2.5	12.0/15.0

Re-drying : 120^{0}C/1hr

Cast Iron & Cutting Electrodes

Z-Ni 55

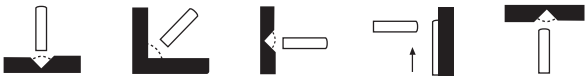
Description and Applications:

- *Basic coated electrode which deposits a machinable nickel iron alloy weld*
- *Designed for strength welding of high duty cast iron such as mechanite, malleable and spheroidal graphite and nodular irons*
- *Welds high phosphorus castings*
- *Typical components to be welded are machine bases, pump bodies, engine blocks, gears and transmission housings*

Specifications:

AWS A5. 15 : E NiFe-CI
DIN 8573 : E NiFe-1-BG-23

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +	~ (Uo : 50V)

Typical all weld metal composition [wt%]

C	Ni	Fe
1.20	55.0	Bal.

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Hardness
-	450	10	200HB

Materials to be welded

Castings, pump bodies, machine bases, engine blocks, gears, cylinders and valves

Lamellar grey cast iron	DIN 1691	GG 10 to GG 35
Malleable cast iron	DIN 1692	GTS 35 10 to GTS 55 4, GTW 35 04 to GTW 45 07, GTW S 38 12
Nodular cast iron	DIN 1693	GGG 40 to GGG 60

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	50-80	2.0	2.0/2.5	12.0/15.0
1/8	3.25	14	350	80-110	3.3	2.0/2.5	12.0/15.0
5/32	4.0	14	350	110-150	5.1	2.0/2.5	12.0/15.0
3/16	5.0	14	350	150-190	7.4	2.0/2.5	12.0/15.0

Re-drying : 100-150°C/1-2hr

Cast Iron & Cutting Electrodes

Z-GSM

Description and Applications:

- *Basic coated electrode for hot welding of nodular cast iron*
- *Suitable for welding malleable black and grey cast iron*
- *Non machinable weld deposit*

Specifications:

DIN 8573 : E FeC-G-BG-26

Welding position

ASME:	1G	2F	2G	3G
ISO 6947	PA	PB	PC	PF

Current

= +	~ (Uo : 70V)

Typical all weld metal composition [wt%]

C	Si	Fe	Mn	Ni
1.00	0.80	3.60	1.30	Bal.

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Hardness
-	520	10	200HB

Materials to be welded

Malleable cast iron	DIN 1692	GTS 35, GTS 45
Nodular cast iron	DIN 1693	GGG 40 to GGG 50

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	60-90	1.7	5.0	15.0
1/8	3.25	14	350	90-120	2.7	5.0	15.0
5/32	4.0	14	350	130-180	4.1	5.0	15.0

Re-drying : 300°C/2h

Cast Iron & Cutting Electrodes

Z-GM

Description and Applications:

- *Monel type core wired electrode for welding lamellar grey, nodular and malleable cast iron*
- *Suitable also for welding of cast iron to mild steel*
- *Machinable weld deposit*

Specifications:

AWS/ASME SFA A5. 15 : ENiCu-B
DIN 8573 : E NiCu-BG-23

Welding position

ASME:	1G	2F	2G	3G
ISO 6947	PA	PB	PC	PF

Current

= +	~ (Uo : 50V)

Typical all weld metal composition [wt%]

C	Fe	Cu	Ni
0.40	2.60	33.00	Bal.

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Hardness
-	400	8	150HB

Materials to be welded

Lamellar grey cast iron	DIN 1691	GG 10
Malleable cast iron	DIN 1692	GTS 35
Nodular cast iron	DIN 1693	GGG 40

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	50-80	1.8	2.0/2.5	12.0/15.0
1/8	3.25	14	350	80-110	3.2	2.0/2.5	12.0/15.0
5/32	4.0	14	350	110-150	4.9	2.0/2.5	12.0/15.0
3/16	5.0	14	350	150-190	7.4	2.0/2.5	12.0/15.0

Re-drying : 120°C/1-2h

Cast Iron & Cutting Electrodes

Z-71

Description and Applications:

- *Electrode for scrapping and gouging*
- *Usable in all positions on mild and low alloy steels, stainless steels, cast iron, nickel based alloys and aluminium alloys*

Working position

All welding positions

Current

= +	~ (Uo : 50V)

Welding parameters and packaging data

Dia		Length		Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
in	mm	in	mm				
3/32	2.5	14	350	120 Min	1.9	3.5	10.5
1/8	3.25	14	350	170 Min	3.3	3.5	10.5
5/32	4.0	14	350	190 Mn	4.9	3.5	10.5
3/16	5.0	14	350	275 Min	7.7	3.5	10.5
1/4	6.0	14	350	350 Min	9.8	3.5	10.5

Cast Iron & Cutting Electrodes

Z-72

Description and Applications:

- *Electrode for cutting and piercing*
- *Usable in all positions on mild and low alloy steels, stainless steels, cast iron, nickel based alloys and aluminium alloys*

Working position

All welding positions

Current

= +	~ (Uo : 50V)

Welding parameters and packaging data

Dia in	Dia mm	Length in	Length mm	Current [amps]	Weight kg/100 pcs	Weight packet [kg]	Weight carton [kg]
3/32	2.5	14	350	120 Min	1.6	4.5	13.5
1/8	3.25	14	350	170 Min	3.7	4.5	13.5
5/32	4.0	14	350	190 Min	5.4	4.5	13.5
3/16	5.0	14	350	275 Min	7.8	4.5	13.5

Z-OXYCUT

Description and Applications:

- *Flux coated tubular electrode for cutting any material by means of oxygen pressure*

Working position

All welding positions

Current

= +

Welding parameters and packaging data

Dia		Length		Current [amps]
in	mm	in	mm	
3/16	5.0	18	450	200-250
9/32	7.0	18/39.3	450/1000	250-300

Flux Cored Wire

Z-71T1

Description and Applications:

- *Rutile Flux cored wire suitable for all-positional welding*
- *Stable and penetrative arc with little spatter formation*
- *High impact toughness at sub-zero temperatures*

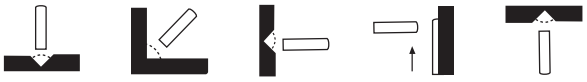
Specifications:

AWS/ASME SFA A5.20 : E71T-1

Approvals

ABS	LR
3YSH5	3YSH5

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

Typical all weld metal composition [wt%]

C	Mn	Si
0.06	1.45	0.45

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
540	580	26	110 J @ -20°C	U

Materials to be welded

Non alloyed sturctural steels	DIN 17 100	St 33, St 37.2, St 44.2, St 52.3
Boiler plates	DIN 17 155	HI, HII, 17 Mn 4
Pipe steels	DIN 1626/1629	St 37.0 to St 52.0
	DIN 1628/1630	St 37.4 to St 52.4
	DIN 17 175	St 35.8, St 45.8
	DIN 17 172	St E 210.7 to St E 415.7
	DIN 17 172	St E 290.7 TM to St E 480.7 TM
	API 5LX	X42 to X46
Fine grained structural steels		St E 255 to St E 460
Shipbuilding steels		A, B, D, E

Packaging data

Dia in	Dia mm	Weight/reel 300 D Outside Dia
0.045	1.2	15 .0
0.063	1.6	15.0

Shielding gas: CO_2

Flux Cored Wire

Z-71T5

Description and Applications:

- *Low hydrogen basic flux cored wire suitable for all-positional welding*
- *Stable and penetrative arc with little spatter formation*
- *Extra high impact toughness at sub-zero temperatures*

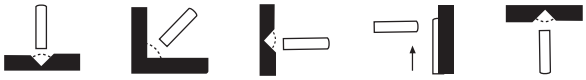
Specifications:

AWS/ASME SFA A5.20 : E71T-5

Approvals

LR
3YSH5

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

Typical all weld metal composition [wt%]

C	Mn	Si
0.06	1.30	0.38

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
480	550	27	100 J @ -30°C	U

Materials to be welded

Non alloyed sturctural steels	DIN 17 100	St 33, St 37, St 44, St 52, St 50, St 60, St 70
Boiler plates	DIN 17 155	HI, HII, 17 Mn 4, 19 Mn 5
Pipe steels	DIN 1626/1629	St 37.0, St 37.4, St 44.0, St 44.4 St 52.0, St 52.4, St 35.8, St 45.8, St E 210.7, ST E 240.7, St E 290.7, St E 320.7, St E 360.7 including thermo-mechanical grades
Fine grained structural steels	DIN 17 102	St E 255 to St E 355, W St E 255 to W St E 355, T St E 255 to T St E 255
Shipbuilding steels		A, B, D, E
Cast steels		GS 38, GS 52, GS 60

Packaging data

Dia in	Dia mm	Weight/reel 300 D Outside Dia
0.045	1.2	15 .0
0.063	1.6	15.0

Shielding gas: CO_2

SEMI AUTOMATIC SOLID WIRE

Z-R2

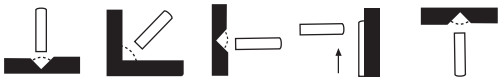
Description and Applications:

- *Double de-oxide mild steel mig solid wire for welding of unalloyed steels and fine grained C-Mn steels*
- *Capable of producing high quality welds where the steel is rusty*

Specifications:

AWS/ASME SFA A5.18 : ER-70S-2
EN 440 : G2 Ti

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

= +

Typical all weld metal composition [wt%]

C	Mn	Si	Ti	Zr	Al
0.06	1.10	0.50	0.10	0.09	0.10

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
440	520	28	100 J @ -30°C	U

Materials to be welded

Unalloyed construction steels	DIN 17 100	St 33 to St52.3
Boiler plates	DIN 17 155	HI, HII , 17 Mn 4
Pipe steels	DIN 1629	St35 to St52.4
Shipbuilding steels		A, B, D, E

Packaging data

Dia in	Dia mm	Weight/reel 300 D Outside Dia	Weight/reel 200 D Outside Dia
0.024	0.6	15 Kgs	5Kgs
0.030	0.8	15 Kgs	5Kgs
0.035	0.9	15 Kgs	5Kgs
0.039	1.0	15 Kgs	5Kgs
0.045	1.2	15 Kgs	5Kgs
0.063	1.6	15 Kgs	5Kgs

Shielding gas: Argon CO_2 (M21)

SEMI AUTOMATIC SOLID WIRE

Z-R3

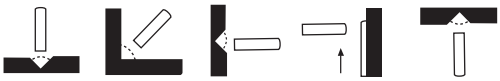
Description and Applications:

- *Double de-oxide mild steel mig solid wire for welding of unalloyed steels and fine grained C-Mn steels*

Specifications:

AWS/ASME SFA A5.18 : ER-70S-3
DIN 8575 : SG 1
EN 440 : G2 Si

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

Typical all weld metal composition [wt%]

C	Mn	Si
0.07	1.20	0.60

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
420	530	28	130 J @ -20°C	U

Materials to be welded

Unalloyed construction steels	DIN 17 100	St 33 to St52.3
Boiler plates	DIN 17 155	HI, HII , 17 Mn 4
Pipe steels	DIN 17 172	StE 210.7 to StE 310.7
	DIN 1629	St35 to St52.4
Shipbuilding steels		A, B, D, E

Packaging data

Dia in	Dia mm	Weight/reel 300 D Outside Dia	Weight/reel 200 D Outside Dia
0.024	0.6	15 Kgs	5Kgs
0.030	0.8	15 Kgs	5Kgs
0.035	0.9	15 Kgs	5Kgs
0.039	1.0	15 Kgs	5Kgs
0.045	1.2	15 Kgs	5Kgs
0.063	1.6	15 Kgs	5Kgs

Shielding gas: Argon CO_2 (M21)

SEMI AUTOMATIC SOLID WIRE

Z-R6

Description and Applications:

- *Double de-oxide mild steel mig solid wire for welding of unalloyed steels and fine grained C-Mn steels*
- *Produces a quality weld with excellent radiographic and mechanical properties*

Specifications:

AWS/ASME SFA A5.18 : ER-70S-6
DIN 8559 : SG 2
BS 2901 : A18
EN 440 : G3 Si 1 mod

Approvals

LR
3YSH15

Welding position

ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Typical all weld metal composition [wt%]

C	Mn	Si
0.08	1.50	0.90

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
480	580	27	60 J @ -30°C	U

Materials to be welded

Unalloyed construction steels	DIN 17 100	St 33 to St52.3
Boiler plates	DIN 17 155	HI, HII , 17 Mn 4
Pipe steels	DIN 17 172	StE 210.7 to StE 360.7
	DIN 1629	St35 to St52.4
Shipbuilding steels		A, B, D, E

Packaging data

Dia in	Dia mm	Weight/reel 300 D Outside Dia	Weight/reel 200 D Outside Dia
0.024	0.6	15 Kgs	5Kgs
0.030	0.8	15 Kgs	5Kgs
0.035	0.9	15 Kgs	5Kgs
0.039	1.0	15 Kgs	5Kgs
0.045	1.2	15 Kgs	5Kgs
0.063	1.6	15 Kgs	5Kgs

Shielding gas: Argon CO_2 (M21)

SEMI AUTOMATIC SOLID WIRE

Z-RMo

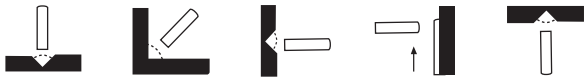
Description and Applications:

- *Molybdenum mild steel wire*
- *Recommended for welding mild steels or 0.5% Mo alloyed steels for high temperatures service*
- *Ideal for avoiding porosity and cracks*

Specifications:

AWS/ASME SFA A5.28 : ER-80S-D2
DIN 8559 : SG Mo
BS 2901 : A 30/A31
EN 440 : G4 Mo

Welding position



ASME:	1G	2F	2G	3G	4G
ISO 6947	PA	PB	PC	PF	PE

Current

Typical all weld metal composition [wt%]

C	Mn	Si	Mo
0.08	1.90	0.70	0.50

Typical all weld metal mechanical properties

Rp 0.2 [MPa]	Rm [MPa]	A_5 [%]	Impact [ISO-V]	Heat Treatment
500	620	25	80 J @ -30°C	U

Materials to be welded

Cast iron	DIN 17 245	GS 22 Mo 4
Boiler plate	DIN 17 155	15 Mo 4

Packaging data

Dia in	Dia mm	Weight/reel 300 D Outside Dia	Weight/reel 200 D Outside Dia
0.030	0.8	15 Kgs	10.5
0.035	0.9	15 Kgs	10.5
0.039	1.0	15 Kgs	10.5
0.045	1.2	15 Kgs	10.5
0.063	1.6	15 Kgs	10.5

Shielding gas: Argon CO_2 (M21)